



COLORADO MINED LAND RECLAMATION BOARD

**APPLICATION FOR
REGULAR 112 PERMIT
DOWE FLATS MINE**

CEMEX, Inc.

**SHB AGRA Project E92-7075
June 1, 1993**

ORIGINAL

RECEIVED

REGULAR OPERATION (112)
RECLAMATION PERMIT
APPLICATION FORM

JUN 1 1993

Division of Minerals & Geology

The application for a Regular Operation Reclamation Permit contains three major parts: (1) the application form; (2) Exhibits A-S; and, (3) the application fee. When you submit your application, be sure to include five (5) signed copies of the application form, five (5) copies of Exhibits A-S, and a check for the application fee described under (3) below. To expedite processing, please provide the information in the format and order described in this form.

M-93-41

Recd 23835 + 23847

Rec 4325 6/2/93

GENERAL OPERATION INFORMATION

Type or print clearly, in the space provided, all information described below.

1. Applicant/operator name: Southdown, Inc. dba Southwestern Portland Cement
 - 1.1 Type of organization (corporation, partnership, etc.): Corporation
2. Operation name (pit, mine or site name): Dowe Flats Mine
 - 2.1 I.R.S. Tax ID No. or Social Security No.: [REDACTED]
3. Application fee for permitted acreage* 1,911 permitted acres
(\$50.00 ÷ \$15.00/acre, 1st. 50 acres;
\$10.00/acre, 2nd. 50 acres; \$5.00/acre, 3rd. 50 acres;
\$1.00/acre, any additional acreage; maximum fee of \$2,000.00)* \$2,325.00
Note permitted acreage and application fee and attach a check for this amount when you file. \$2,000.00 application fee
4. Substance(s) to be mined (e.g., gravel/gold/borrow): Limestone and Shale Quarry
5. Name of owner of surface of affected land+ Refer to Exhibit P
6. Name of owner of subsurface rights of affected land+ Southdown, Inc. and Harroun
subject to Southdown, Inc.'s
option rights
7. Type of mining operation: ☒ Surface ☐ Underground ☐ In-situ
8. Location Information: the center of the area where the majority of mining will occur lies in:
COUNTY: Boulder

PRINCIPAL MERIDIAN (check one): ☒ 6th (Colorado) ☐ 10th (New Mexico) ☐ Ute
TOWNSHIP (write number and check direction): T 3 ☒ North ☐ South
RANGE (write number and check direction): R 70 ☐ East ☒ West
SECTION (write number): S 9, 10, 15, 16, 21, 22
QUARTER SECTION (check one): ☐ NE ☐ NW ☐ SE ☐ SW
QUARTER/QUARTER SECTION (check one): ☐ NE ☐ NW ☐ SE ☐ SW
GENERAL DESCRIPTION:
(miles and direction from nearest town and approximate elevation) Approximately 2 miles east of Lyons, Colorado
Approximate elevation 5,350 feet

- * If the operator is a unit of state or county government, an application fee is not required. A fraction of an acre must be counted as one acre for fee calculation.
+ If 2 or more, refer to Exhibit P

9. Correspondence Information:

APPLICANT/OPERATOR (name, address, and phone of name to be used on permit)

Individual's Name: Gerald Anderson or John Lohr
Company Name: Southdown, Inc.
Street: P.O. Box 529
City: Lyons
State: Colorado Zip Code: 80540
Area Code: 303 Telephone: 523-4206

PERMITTING CONTACT (if different from applicant/operator above)

Individual's Name: Above or Paul Banks
Company Name: SHB AGRA, Inc.
Street: 165 So. Union Blvd., Suite 516
City: Lakewood
State: Colorado Zip Code: 80228
Area Code: 303 Telephone: 763-8432

INSPECTION CONTACT

Individual's Name: Gerald Anderson or John Lohr
Company Name: Southdown, Inc.
Street: P.O. Box 529
City: Lyons
State: Colorado Zip Code: 80540
Area Code: 303 Telephone: 523-4206

CC: STATE OR FEDERAL LANDOWNER (if any)

Agency: None
Street: _____
City: _____
State: _____ Zip Code: _____
Area Code: _____ Telephone: _____

CC: STATE OR FEDERAL LANDOWNER (if any)

Agency: _____
Street: _____
City: _____
State: _____ Zip Code: _____
Area Code: _____ Telephone: _____

10. Primary future (Post-mining) land use (check one):
- | | | |
|--|--|--|
| ☐ Cropland(CR) | ☐ Pastureland(PL) | ☐ General Agriculture(GA) |
| ☐ Rangeland(RL) | ☐ Forestry(FR) | ☒ Wildlife Habitat(WL) |
| ☐ Residential(RS) | ☐ Recreation(RC) | ☐ Industrial/Commercial(IC) |
| ☐ Developed Water Resources(WR) | | ☐ Solid Waste Disposal(WD) |
11. Primary present land use (check one):
- | | | |
|--|--|--|
| ☒ Cropland(CR) | ☐ Pastureland(PL) | ☐ General Agriculture(GA) |
| ☐ Rangeland(RL) | ☐ Forestry(FR) | ☐ Wildlife Habitat(WL) |
| ☐ Residential(RS) | ☐ Recreation(RC) | ☐ Industrial/Commercial(IC) |
| ☐ Developed Water Resources(WR) | | |

An applicant will not be issued a reclamation permit until notified by the Division that this application and the required performance and financial warranties have been approved. Alternatively, an automatic approval will occur where the Division fails to notify the applicant/operator that the application has been denied. The final agency action must be made within 120 calendar days from the date the application was submitted. However, the performance and financial warranties must be approved by the Division or Board before a permit will be issued even if you receive an automatic approval. NO MINING OPERATIONS SHALL BEGIN UNTIL A PERMIT IS ISSUED (Rule 7.2(2)).

It shall be a provision of this permit that the operations will be conducted in accordance with the terms and conditions listed in your application, as well as with the provisions of the Act and the Mineral Rules and Regulations in effect at the time the permit is issued.

Notice Requirements

Immediately after the first publication of your notice in a newspaper of general circulation in the locality of your proposed mining operation, you must mail a copy of the notice to all owners of record of surface rights to the affected land and all owners of record of lands that are within 200 feet of the boundary of the affected land. Notice requirements are specified in Rule 2.2. A copy of a form which includes all required information for the notice has been attached for your use. You will need to provide the Division proof of notice prior to the decision date, which may be by submitting return receipts of a certified mailing or by proof of personal service (Rule 2.2.2(6)).

NOTE TO COMMENTORS/OBJECTORS: It is likely there will be additions, changes, and deletions to this document prior to final decision by the Mined Land Reclamation Board or Division. Therefore, if you have any comments or concerns, you should contact the applicant or the Division prior to the decision date so that you will know what changes may have been made to the application document.

The Division is not allowed to consider comments, unless they are written, and received prior to the end of the public comment period. You should contact the applicant for the final date of the public comment period.

If you have questions about the Mined Land Reclamation Board and Division's review and decision or hearing process, you should contact the Division at (303) 866-3567.

As an authorized representative of the applicant, I hereby certify that the operation described herein has met the minimum requirements of the following terms and conditions:

1. This mining operation will not adversely affect the stability of any significant, valuable and permanent man-made structure(s) located within two hundred (200) feet of the affected lands. However, where there is an agreement between the applicant/operator and the persons having an interest in the structure that damage to the structure is to be compensated for by the applicant/operator (C.R.S. 34-32-115(4)(d)), then mining may occur within 200 feet. Proof of an agreement must be submitted to the Division prior to the decision date.
2. No mining operation will be located on lands where such operations are prohibited by law (C.R.S. 34-32-115(4)(f));
3. As the applicant/operator, I do not have any mining/prospecting operations in this State of Colorado currently in violation of the provisions of the Mined Land Reclamation Act (C.R.S. 34-32-120) as determined through a Board finding.
4. I understand that statements in the application are being made under penalty of perjury and that false statements made herein are punishable as a Class 1 misdemeanor pursuant to Section 18-6-503, C.R.S.

Signed and dated this 20th day of May, 93.

Applicant/Operator

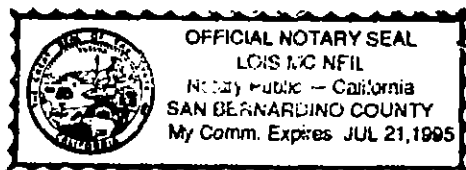
By: Michael A. Yannone
Title: _____

If Corporation Attest (seal)

By: Wade E. Kelly
Corporate Secretary or Equivalent
Town/City/County Clerk

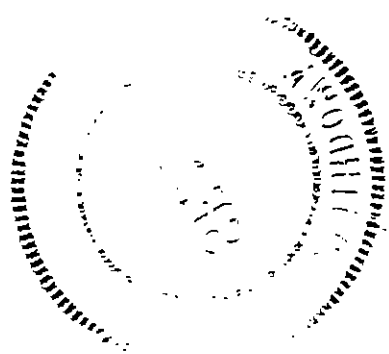
State of California)
County of San Bernardino) ss.

The foregoing instrument was acknowledged before me this 20 day of May, 93,
by Michael A. Yannone as Dir., Proj. & Services of Southdown, Inc.
dba Southwestern Portland Cement



Lois McNeil
Notary Public

My Commission expires: _____



**DOWE FLATS MINE
COLORADO MINED LAND RECLAMATION BOARD
REGULAR 112 PERMIT APPLICATION**

GENERAL INFORMATION

Name of Operation: Dowe Flats

Name of Applicant: Southdown, Inc. dba Southwestern Portland Cement

Address and Phone Number: Southdown, Inc.
P.O. Box 529
Lyons, CO 80540
(303) 534-4206

Parent Corporation: Southdown, Inc.
1200 Smith Street, Suite 2200
Houston, Texas 77002

Surface Owner: See Exhibit P

Mineral Owner: See Exhibit P

Source of Legal Right to Enter: See Exhibit O

Type of Mine and Substance to be mined: Open pit limestone and shale mine

Present Land Use of Area: Agricultural

Surrounding Land Uses: Agricultural, Low Density Residential, cement plant, limestone quarries, industrial, open land, range land.

Proposed Future Land Use: Wildlife habitat

34-32-115(4)(f)C.R.S.: The lands described herein do not affect any of the lands described in this section of State statute.

EXHIBIT A

LEGAL DESCRIPTION

**PARTS OF SECTIONS 9, 10, 15, 16, 21, AND 22
TOWNSHIP 3 NORTH, RANGE 70 WEST 6TH P.M.**

SEE MAP EXHIBIT A

✓ Rule 2.1.2 (2)
legal description must
be provided to at least
the nearest quarter
quarter section.
Please provide
this

Exhibit C-1
Pre-Mining Maps
Please make a list of all man-made structures within 200 ft. of the affected boundary

EXHIBIT D - MINING PLAN

The Division will require a letter from the owner of the mine to the operator of each of these features which stipulates who is responsible for any damage to the permanent or man-made structure as a result of the mining operation

Rule 21.12 (4) (b) (g)
Show name & location of all permanent structures or affected area within 200 ft. of affected area

Four horizons have been targeted for mining at Dowe Flats based principally on their calcium carbonate content and utilization potential at Southdown's nearby cement plant. The west limb of a southward plunging syncline contains the four beds of interest; all outcrop along a general north/south strike and exhibit gentle dips to the east. The Western most seam (also the stratigraphically lowest) has been somewhat resistant to erosion and is referred to as the Hi-Cal ridge. Expressions of the other horizons are more subdued and these are known locally as simply the 2nd, 3rd, and 4th ridges (Map C-2 shows the areal extent of mining for the four ridges with the overlaps due to the stratigraphic superposition of the limestone beds).

A primary objective of the mining plan has been to achieve a 25-year project life, given the expected product mix deriving from the four minable seams and the overall annual tonnage that must be delivered to the cement plant. Currently this annual figure is 760,000 tons, with roughly 40% to 45% of this amount contributed each by the Hi-Cal and 3rd ridges. Because the Hi-Cal ridge, at its outcrop, dips more steeply than the others, the desire is to maximize mining along its strike length and to avoid excessive down-dip extraction; the objective is to minimize waste handling associated with the removal of the limestone beds. As a result of these factors, it has been determined that the Hi-Cal bed will need to be mined over a strike length of almost 8,000 feet and that the 2nd and 3rd seams will be extracted over a linear distance of 5,000 feet. Only 1,900 feet of the 4th ridge outcrop will be disturbed.

It is stated in the app that Southdown Prefers to mine in 100 acre phases (Are all 4 pits active at one time?)

(d) boundary of area of affected lands or boundary of entire area to be involved in the operation
This includes within 200 ft. of the haulroad

Mining of each of the respective beds is planned to begin at a southern point along the outcrop, with the pit faces advancing toward the north. Since the limestone will be transported to the south for delivery to the plant, the southerly starting location should serve to minimize haulage costs during the early years of operation.

Please provide a description and a detailed map showing the 100 ac phases and what portion of each phase is excavation, stockpile area, storage area, etc.

Entry into each of the four pits will be from the west; in-pit and mine haulroads are planned with a surface width of 60 feet in order to allow safe two-way traffic, and grades will be held to a maximum of 8%. This latter figure will be significant only on the Hi-Cal ridge wherein the outcrop dip attains an inclination of nearly 13%.

The initial activity at the site will entail the stripping and stockpiling of topsoil taken from each of the four starter pits as well as from the planned locations of the permanent out-of-pit waste storage piles. This will be followed by removal of overburden material from the targeted limestone beds and placement of this waste in locations consistent with achieving the post mining contour shown in the reclamation plan. Exposed limestone will then be extracted and delivered to the nearby plant.

Give an approx. timetable for these phases & describe how reclamation fits in with schedule

C-2 show mining phases (including topsoil stockpiles, stormwater, control barriers, ditches, check dams, etc.)

* estimate of the volume of topsoil overburden & pit

show direction of mining on Exh C-2

After sufficient forward progress has been attained so that equipment can move freely in the pits, all waste material removed from the advancing face in each of the four horizons will be used in sculpting the ground surface to final form in accordance with the approved reclamation plan. Backfilling and/or reshaping of the pits will then continue for the remainder of the project life. Once sculpting has been initiated, the waste material will be levelled into a permanent landform, topsoil replacement will begin, and revegetation will be started. These activities will continue throughout the project life.

All field equipment will be diesel powered and no powerlines will need to be constructed into Dowe Flats. Drills will be required to prepare the overburden and limestone for blasting; dozers are needed to assist in levelling drill sites following removal of any topsoil with scrapers. The blasted rock will be lifted by front loaders and deposited into off-road haulers for disposition onto land to be reclaimed or for transportation to the plant (limestone).

Production rock will be hauled to the plant by truck on a private, dedicated haul road. Trucks will either be 25 ton or 85 ton dumps, depending on the conditions of approval for the Highway Access Permit issued by the Colorado Dept. of Transportation (Southdown's preference is the 85 ton trucks in order to reduce the number of crossings of State Highway 66). The haul road location is shown on Map C-2 and will cross four irrigation ditches, and State Highway 66 and the Burlington Northern Railroad track at grade. The road will go over the St. Vrain River, cross land owned by Frontier Materials, Inc., then cross Boulder County Road 49 and enter the cement plant property.

Improvements to public roadways will be performed as necessary and as required by the Boulder County Public Works Department and the Colorado Department of Transportation.

The Hi-Cal bed is made up of limestone beds 1-2 feet in thickness separated by thin shale or clay partings. This bed has the highest lime content of any of the horizons at Dowe Flats and therefore will be the most important of the horizons mined. In order to obtain sufficient material from this bed to supply plant needs for the 25-year project life, a maximum 1,100-foot width pit is proposed. Maximum pit depth is estimated to be approximately 125 feet as measured from the original ground surface.

The 2nd Ridge consists of many thin limestone seams interbedded with shale. Reserves have been calculated to a pit depth of 70 feet.

SB-181 requires
points of compliance
designate upgradient
& downgradient
groundwater monitoring
points

If so, could be
classified as a Designated
Mining of
with
additional
requirements
to be
met in
environmental
protection
plan

The 3rd Ridge has a thickness of about 30 feet. Recent studies have shown that the formation may be divided into three bands: a top higher-alkali portion nearly 8 feet thick; a center lower-alkali band 15-16 feet in thickness; and a bottom zone of lower lime content which may be upwards of 10 feet thick. These bands have traditionally been designated by chemical analysis. They are very difficult to distinguish in the quarries by visual observation. Maximum pit depths will likely be 60 feet.

Provide
analyses
of the
content
of this rock

will pyritic material be
removed and exposed
to oxidation
(acid
drainage?)

The 4th Ridge formation contains relatively high quantities of pyrite where it is fresh and unaltered. Near-surface weathering has removed the pyrite and has upgraded at least this portion of the 4th Ridge to a useful material. Mining depths in this horizon will be somewhat shallower than in the other limestone units, with a maximum of approximately 30 feet.

will the
base of a
body of water
sit on
pyritic
rock?

The configuration of mining depicted in the drawings with this report has given consideration to the possibility of eventual dam construction at grid 9100N. For dam stability reasons, mining of the HiCal limestone underneath the proposed dam (and south of the dam) has been limited to a maximum pit depth of 65 feet. Beginning at a distance of 300 feet north of the upstream toe of the dam, extraction of this limestone unit will widen to approximately 1,100 feet from the outcrop, at which point the pit floors will be nearly 125 feet from the original ground surface. The stratigraphically higher 2nd ridge will be intersected throughout much of its length by the HiCal pit; therefore prior removal of the 2nd ridge overburden and limestone material will be beneficial in that fewer yards of waste rock will be attributed to HiCal mining.

The 2nd, 3rd, and 4th ridges will not be extracted beneath, or south of, the dam footprint. Mining will begin approximately 200 feet north of the projected upstream dam toe and will advance to the north at a rate commensurate with limestone consumption.

The Harroun farm will be the mine operations center, including an office, maintenance building, sanitary facilities and equipment parking and storage area. Some of the existing structures will be demolished and some new structures will be built. Existing water and sewer on the Harroun farm will be upgraded to serve up to 20 mine employees and visitors. Southdown owns adequate water rights to supply approximately 150,000 gallons per day needed to operate the mine.

A summary of mine data is shown on the attached spreadsheet.

**DOWE FLATS PROJECT - LYONS, CO
MINING & RECLAMATION PLAN SUMMARY**

	Hi-Cal	2nd Ridge	3rd Ridge	4th Ridge	TOTAL
Term in Years	25	25	25	25	25
Production Tons	7,600,000	1,700,000	8,900,000	800,000	19,000,000
Acres*	170	56	104	6	336
Cubic Yards of Production Rock	3,800,000	850,000	4,450,000	400,000	9,500,000
Mine Waste in Cubic Yards**	10,000,000	2,000,000	2,100,000	182,000	14,300,000

* Ridges overlap; acreage given is for each ridge regardless of overlap. total acreage mined is 312.

** Includes topsoil, unconsolidated overburden, and waste rock. Volumes shown are bank cubic yards. Bulking factors of 20% for unconsolidated overburden and 50% for waste rock were used. The total volume of loose cubic yards placed on the reclaimed surface is 20,800,000.

EXHIBIT E - RECLAMATION PLAN

Type of Reclamation

The Dowe Flats reclamation plan has been developed with the following main objectives:

- Create the most diverse wildlife habitat possible; the diverse topography made possible by mining makes this objective particularly attractive. The opportunities to create "new" topography, establish woody vegetation in various parts of the post-mining landscape, and take advantage of water entering the lower parts of excavated areas will greatly enhance the value of the area to a variety of game and non-game wildlife. Specifically, wildlife habitat construction is directed toward 1) establishment of raptor prey base comprised of prairie dogs, lagomorphs, other small mammals, waterfowl, and non-game birds, 2) establishment of habitat consistent with use as mule deer movement corridor and winter range, 3) establishment of raptor roost, perch and/or nest sites, and 4) establishment of riparian habitat useful as feeding, breeding, and refuge habitat for a large variety of mammalian, avian, and herptile species.
- Reduce visual liabilities of the strong north-south linear nature of the mining disturbance by introducing topographic grain at differing angles and providing the distraction of varying vegetational texture and form. This necessitates establishment of cross-pit topography as well as topography extending away from the mine area to connect visually with adjacent natural topographic features, thereby reducing the visual impact of the previous linear mining features underlying the area.
- Construct a mosaic of vegetation that includes species and plant community structures such as naturally occur and will tie in with landscapes of the areas adjacent to Dowe Flats on Rabbit Mountain and Indian Mountain.

In addition, the distribution of topographic features in the reclaimed landscape (Exhibit F) has been developed in consideration of the possible construction of an earthen dam across the area mined as part of this project. Specifically, no reclamation topography has been constructed within the footprint of the dam for the first seven years of mining. After that, reclaimed topographic features extend across part of the dam footprint.

Rule 2.1.2 (b) (e) (i) - (iii)
need (1) estimate of period of time which will be required for various stages or phases of reclamation

(2) Description of the size + location of each area to be reclaimed during each phase

(3) Outline the sequence in which each stage or phase of reclamation will be carried out.

Exhibit F shows the post-mining and reclamation topography. Reclamation will proceed at the earliest stages of mining as topography is built from overburden materials, and topsoil is replaced as shown in Table E-1. Earth will be moved from the area to be mined to its final site by use of the mine haulage trucks, whereupon it will be graded to the desired contours by dozers. Topsoil will be hauled to the site in trucks or scrapers and spread to depths indicated in Table E-1.

Post-reclamation Land Use

Land use after reclamation, in the area that was disturbed by mining and reclamation, will be primarily wildlife habitat and will be consistent with similar uses of adjacent and nearby lands, both public and private. Steep land to the west on Indian Mountain and to the east on Rabbit Mountain is currently valuable wildlife habitat, and the development of diverse topography and vegetation cover on what had formerly been primarily plowed marginal agricultural land will enhance the usefulness of the entire area to wildlife.

Wildlife Habitat Construction

Features of the Dowe Flats Reclamation Plan that address enhancement of wildlife habitat include the following:

Rimrock

Lagomorphs will be encouraged through development of hillside "rimrock" burrow habitat.

Hunting Perches

Hunting by raptors will be encouraged by establishment of hilltop perches, initially telephone pole artificial perches followed by development of cottonwood and pine trees in hilltop/hilltop depressions (Figure E-1)

Ponds/Wetlands

A large variety of birds as well as mammals, reptiles, and amphibians will be encouraged by the creation of mosaic ponds and herbaceous and wooded wetlands.

Prairie Dog Habitat

In the east valley, existing degraded cultivated land will be replaced with shallow soil grassland interspersed with new ridges. Prairie dogs are expected to occupy these vegetated areas without impediment.

Reclamation Schedule

Reclamation will occur concurrently with mining and will proceed, generally, from south to north. Where possible, topsoil will be immediately hauled directly to graded sites and placed prior to revegetation. In some cases, topsoil will need to be stored in the areas indicated on Exhibit F.

Reclamation Procedures

Construction of Topographic Features

Topographic features will be constructed of coarse waste rock. The waste rock core of these features will be covered with 24-60 inches of fine overburden except where "rimrock" structures are planned (Figure E-2). These "rimrock" features will expose linear bands of coarse rock on hillsides and will provide habitat for small mammals, especially hares and rabbits, that will provide part of the prey base of raptors, now dependent largely on a single species (i.e. prairie dogs). Voids within the coarse rock will offer burrow habitat not available elsewhere in the reclamation (or existing) landscape.

Subsoil Placement

In the areas of the excavated limestone and on the surface of waste rock cores of constructed topography, fine overburden will be placed to depths from 2 to 5 feet prior to topsoil placement to provide subsoil for revegetation.

Topsoil Salvage

Topsoil will be salvaged to a depth of 12 inches. Soil will be transported to one of the storage piles indicated on Exhibit F, or placed directly on a graded surface in preparation for seeding/planting.

Topsoil Storage

Topsoil will be stored in piles with slopes no steeper than 4(h):1(v). An interceptor berm will be constructed around each pile. The stored material will be stabilized as indicated under Temporary Surface Stabilization, below.

Topsoil Placement

Topsoil will be replaced on graded slopes at depths indicated on Table E-1. Prior to placement, the graded slopes will be left rough to facilitate adhesion of the topsoil layer. Topsoil depth will be monitored during placement by reclamation personnel to verify that it is within the specified range.

Seedbed Preparation

Please explain ✓ Areas to be seeded will be ripped following topsoil placement to a depth of 18 inches to relieve subsoil compaction caused by heavy machinery traffic. *How many acres to be ripped?* Areas to be drill seeded will be cultipacked; if necessary, they will be diced to reduce clod size. If ready during the period of April 1 to October 1, the appropriate cover crop will be sown (see Mulching below). *Mulch* ✓ If outside that time range, 2 tons/ acre of clean long-fiber straw will be spread and crimped into the surface. Areas to be broadcast seeded will be harrowed immediately prior to and after seeding to assure maximal contact of seed with uncrusted soil and at least moderate seed burial depth.

Seed Mixes

Listed in Tables E-2 through E-9 are planting specifications for each of 8 different post-reclamation vegetation types. In these tables are included both seed mixes and nursery stock planting rates. For the seed mixes, rates of Pure Live Seed (PLS)/acre for drill seeding and broadcast seeding are shown. Plant communities to be created during reclamation include the following:

- 1 • Shallow Soil (Dry) Grassland (Table E-2) - This type is planned for ridgetop sites and slope with southeast to southwest exposures. Topsoil depth will be 4 to 8 inches.
- 2 • Deep Soil (Moist) Grassland (Table E-3) - This type is planned for valley bottom and other low slope areas of varying exposure. The deep soil (topsoil depth 15 to 21 inches) will allow optimal development of grass root systems. Although it is anticipated that moist conditions will prevail, it is not intended for this type to be a wetland community.
- 3 • Toe Slope Grassland (Table E-4) - This type is to be placed at the base of reclaimed slopes below Shallow Soil (Dry) Grassland, Dry Shrubland, or Moist

Shrubland to intercept and contain any sediment from runoff waters passing downslope from these communities from which, in particular, there can be expected to be a more limited ground cover for the short to medium term of vegetation development. Topsoil will be 9 to 15 inches deep.

- 4 • Dry Shrubland (Table E-5) - This type is to be placed on southeast to southwest exposures on shallow soil (topsoil depth 2 to 6 inches).
- 5 • Moist Shrubland (Table E-6) - To be placed on steep to moderately steep northeast to northwest facing slopes on shallow soil (topsoil depth 2 to 6 inches), this type will include a variety of tall shrubs that will provide food, shelter, and breeding habitat to a large variety of mammalian and avian species.
- 6 • Herbaceous Wetland (Table E-7) - This community will be established in low areas that will be wetted by ground water seepage and capillary rise from the water table. Topsoil depth will 9 to 15 inches.
- 7 • Wooded Wetland (Table E-8) - In this community, tall shrubs and trees that tolerate wetland soil conditions will be planted. Substrates in which these will be placed will be 15 to 20 inches of topsoil, with surface elevations set slightly (6 to 12 inches) above water table to allow roots "breathing space." In addition to the species listed, it is anticipated that cottonwoods (*Populus* spp.) will volunteer in various locations both in this and the preceding type. Crack willows will not be planted. ✓
How will this be monitored + assured?
- 8 • Cottonwood Groves (Table E-9) - Areas of cottonwood grove are planned to provide avian (especially raptor) perches as the trees eventually mature. Topsoil depth will be 21 to 27 inches. These areas will be located in concave areas, both in valley bottoms and in upper elevation depressions. To provide the necessary physical features needed for raptor perching before the trees grow tall enough, artificial perches will be provided, consisting of a Wolmanzied pole extending 40 feet above the surface, with a cross-bar at the top, and another located about 5 ft. below. ✓
How do you maintain enough moisture content here?
How recommended?

Seeding Methods

Drill seeding will proceed on slopes of 2(h):1(v) or less steep, oriented on the contour. The seed drill will be a "Grassland" type drill with at least two seed boxes, one of which will have picker wheels, or other means of handling fluffy seed. It will be equipped with double disk furrow openers, depth bands set at 1/4 to 1/2 inch, and packer wheels or drag chains. During planting, the legume, smooth, or small seed box will be filled with the aggressive cool season species indicated in the Plant Material Mixes (Tables E-2 through E-9) and every second and third drop tube out of this box will be blocked so that these aggressive species are placed only in every third furrow; thus, in two furrows of each three, the less aggressive cool season grasses, as well as the warm season grasses and the forbs (and shrubs, where applicable) will be planted alone. This will allow these less aggressive species to develop without the often fatal direct competition of more aggressive cool season grasses. (Note that these listed mixes do not include any of the extremely aggressive introduced cool season grasses that so often overwhelm all other components, even in seed mixes that seem to contain a large proportion of native species).

Drill
Seed
Twice

Double the seeding rate? ✓

Broadcast seeding will be undertaken on slopes steeper than 2(h):1(v). Seeding rates for broadcast are higher than for drill seeding because some substantial portion of seed sown by broadcast fall in microsites from which successful germination and establishment are impossible. Immediately prior to broadcast, the surface will be ^{before} harrowed to expose freshly disturbed soil, which will result in optimal soil-seed contact; immediately following seeding, the surface will be lightly ~~harrowed or raked~~ ^{and after} to cover the seeds lightly.

Mulching

The preferred means of providing the protective cover necessary to shelter germinating seeds is use of a cover crop. Cover crop species and periods during which they may be profitably planted are listed in Table E-10. Use of cover crop is preferred because 1) it results in a stronger vertical surface structure than other mulching methods, providing much superior protection from erosive and desiccating effects of wind, 2) it provides more organic matter and often better surface cover than other forms of mulch, and 3) it reduces the possibility of introduction of additional weed seed. These cover crops will be sown using a drill equipped with coulter wheels and furrow openers spaced between 10 and 16 inches. If the seedbed is not ready to plant between April 1 and October 1, then use of a straw mulch will be undertaken. Two (2) tons of clean long-fiber straw will be spread and crimped into the soil surface prior to

hydromulch
what acreage will be broadcast seeded or hydromulched w/ a cover crop.
seeding. In areas to be broadcast seeded, seeding will precede application of mulch which will be either hydromulched with 1600 lbs. of thermally produced wood fiber mulch, or covered with 2 tons of clean long-fiber straw. In either case, the applied mulch will be anchored with 100 lbs/acre of Psyllium-based tackifier.

Temporary Surface Stabilization

show these on a map + give estimated volume
Certain areas, such as topsoil stockpiles, roadcuts, or fills, will require establishment of a vegetational cover to protect the surface from erosion until disposition of the underlying material in final reclamation. Following completion of earthmoving, such temporary areas will be drill seeded with the seed mix indicated in Table E-11. After seeding, the area will be mulched as indicated in **Mulching** above, either with straw or hydraulically-applied wood fiber, and anchored with tackifier.

Fertilizer

Phosphate in the form of treble superphosphate will be spread at the rate of 30 lb P_2O_5 per acre during seedbed preparation and incorporated into the upper 6 inches of soil. Nitrogen as a slow-release form will be applied at the rate of 20 lbs. N per acre in the same fashion. After the preparation of the seedbed, no fertilizer amendments to permanently seeded areas are planned; such well-intentioned actions have proven to encourage the competition of weeds to the substantial detriment of desirable but often slow-growing native species.

Weed Control

Weed growth in the planted areas will be monitored and any development of weeds that may cause significant damage to the desirable planted species will be discouraged by mowing; or in the case of perennial plants listed as noxious in Colorado and Boulder County, plowed and replanted or treated with a glyphosate-based herbicide. *composed of ? toxic ?* ✓

Control of Wild Browsers

One of the major potential problems that must be controlled in efforts to create wildlife habitat is protection of the young shrubs and trees from the depredation of some of the wildlife in whose benefit the eventual establishment of those plants rests. Units of reclaimed land will be enclosed with 8 foot-high fence that will remain in place for a period of 5 years after the planting of that area. This will allow the shrubs and trees to attain sufficient growth, above and below ground to withstand browsing pressure. *How big? Show on a map.* ✓

Shrub and Tree Planting

Woody plants are to be planted in the reclaimed areas using containerized nursery stock. These shrubs and trees will be planted at rates listed in the Tables E-2, E-4, E-5, E-6, and E-8. Wetland stock listed in Table E-7 will be planted as rhizomes or corms. Trees listed in Table E-9, will be planted as nursery stock in the calipers footnoted there.

Table E-1. Topsoil Placement Depths

<u>Community Type</u>	<u>Avg. Depth (in.)</u>	<u>Acres</u>	<u>Volume (cy)</u>
Shallow Soil (Dry) Grassland	6	160	129,000
Shallow Soil Grassland with Pines	6	49	39,500
Deep Soil (Moist) Grassland	18	10	24,200
Toe Slope Grassland	12	72	116,000
Dry Shrubland	4	227	120,700
Moist Shrubland	4	95	50,500
Herbaceous Wetland	18	33	79,800
Wooded Wetland	12	3	4,800
Cottonwood Groves	24	12	38,700
Water	0	44	0
	Total	<u>705</u>	<u>603,200</u>

Note: Exhibit E shows the location and acreage of each vegetation community type.

Table E-2. Plant Material Mixes - Shallow Soil (Dry) Grassland

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	<u>Seeding</u> <u>Rate (PLS lb/ac)^b</u>	
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Agropyron dasystachyum ^d	Thickspike Wheatgrass - Critana	0.4	1.0
Agropyron inerme ^d	Beardless Bluebunch Whtgs - Whitmar	2.2	5.5
Agropuron riparium ^d	Streambank Wheatgrass - Sodar	0.6	1.5
Agropyron smithii ^d	Western Wheatgrass - Arriba	1.5	3.8
Agropyron spicatum ^d	Bluebunch Wheatgrass - Secar	1.9	4.8
Bouteloua curtipendula	Sideoats Grama - Vaughn	0.9	2.3
Bouteloua gracilis	Blue Grama - Native, Alma	0.2	0.5
Buchloe dactyloides	Buffalo Grass	2.9	7.3
Festuca arizonica ^d	Arizona Fescue - Redondo	0.6	1.5
Koeleria macrantha	Junegrass	0.03	0.08
Oryzopsis hymenoides	Indian Ricegrass - Nezpar, Paloma	1.9	4.8
Poa canbyi	Canby Bluegrass - Canbar	0.2	0.5
Schizachyrium scoparium	Little Bluestem - Blaze, Pastura	1.0	2.5
Stipa comata	Needleandthread	1.4	3.5
Stipa viridula	Green Needlegrass - Lodorm	0.9	2.3
<u>FORBS</u>			
Achillea lanulosa	Western Yarrow	0.01	0.03
Aster glaucodes	Glaucous Aster	0.06	0.15
Coreopsis tinctoria	Plains Coreopsis	0.03	0.08
Heliomeris multiflora	Showy Goldeneye	0.03	0.08
Linum lewisii	Blue Flax	0.11	0.28
Medicago sativa	Alfalfa - Ladak	0.16	0.40
Penstemon palmeri	Palmer Penstemon - Cedar	0.05	0.13
Petalostemon purpureum	Purple Prairie Clover - Kanab	0.11	0.28
Ratibida columnifera	Prairie Coneflower	0.03	0.08
Sphaeralcea coccinea	Orange Globemallow	0.06	0.15
<u>SHRUBS</u>			
Atriplex canescens	Fourwing Saltbush	1.3	3.3
Cercocarpus montanus	Birch-leaf Mountain Mahogany	1.2	3.0
Chrysothamnus nauseosus	Rubber Rabbitbrush	0.2	0.5
		<u>Planting Rate</u>	
		<u>Stems / acre</u>	
Cercocarpus montanus	Birch-leaf Mountain Mahogany	---400----	
Chrysothamnus nauseosus	Rubber Rabbitbrush	---100----	

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded

^d More aggressive species that will be sown only in every third drill furrow.

Table E-3. Plant Material Mixes - Deep Soil (Moist) Grassland

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	Seeding	
		Rate (PLS lb/ac)^b	
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Agropyron dasystachyum ^d	Thickspike Wheatgrass - Critana	0.7	1.8
Agropyron riparium ^d	Streambank Wheatgrass - Sodar	0.7	1.8
Agropyron smithii ^d	Western wheatgrass - Arriba	2.0	5.0
Agropyron trachycaulum ^d	Slender Wheatgrass - San Luis	0.7	1.8
Andropogon gerardii	Big Bluestem - Champ, Native	2.0	5.0
Elymus cinereus ^d	Basin Wildrye - Magnar	0.5	1.3
Festuca rubra	Creeping Red Fescue - Dawson	0.1	0.3
Panicum virgatum	Switchgrass - Nebraska 28	0.6	1.5
Poa ampla	Big Bluegrass - Sherman	0.1	0.3
Poa compressa	Canada Bluegrass - Reubens	0.02	0.05
Sorghastrum nutans	Indiangrass - Holt	1.2	3.0
Stipa viridula	Green Needlegrass - Lodorm	1.2	3.0
<u>FORBS</u>			
Achillea lanulosa	Western Yarrow	0.04	0.10
Aster adscendens	Pacific Aster	0.04	0.10
Gaillardia aristata	Blanketflower	0.33	0.83
Hedysarum boreale	Sweetvetch	0.30	0.75
Linum lewisii	Blue Flax	0.15	0.38
Medicago sativa	Alfalfa - Ladak	0.13	0.33
Monarda fistulosa	Horsemint	0.03	0.08
Penstemon strictus	Rocky Mountain Penstemon - Bandera	0.07	0.18
Solidago rigida	Stiff Goldenrod	0.06	0.15
Thermopsis divaricarpa	Goldenbanner	0.40	1.00

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded

^d More aggressive species that will be sown only in every third drill furrow.

Table E-4. Plant Material Mixes - Toe Slope Grassland

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	<u>Seeding</u> <u>Rate (PLS lb/ac)^b</u>	
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Agropyron dasystachyum	Thickspike Wheatgrass - Critana	2.1	5.3
Agropyron riparium	Streambank Wheatgrass - Sodar	2.1	5.3
Agropyron smithii	Western Wheatgrass - Arriba	4.6	11.5
Andropogon gerardii	Big Bluestem - Champ or Pawnee	2.5	6.3
Elymus cinereus	Basin Wildrye - Magnar	2.5	6.3
Elymus triticoides	Creeping Wildrye - Shoshoni	6.4	16.0
<u>FORBS</u>			
Medicago sativa	Alfalfa - Ladak	0.2	0.5
<u>SHRUBS</u>			
Rhus glabra v. cismontana	Smooth Sumac	<u>Planting Rate</u> <u>Stems / acre</u> ---400---	

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded

Table E-5. Plant Material Mixes - Dry Shrubland

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	Seeding	
		Rate (PLS lb/ac)^b	
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Agropyron inerme ^d	Beardless Bluebunch Whtgs. - Whitmar	2.3	5.8
Bouteloua curtipendula	Sideoats Grama - Vaughn	1.7	4.3
Bouteloua gracilis	Blue Grama - Alma	0.4	1.0
Festuca arizonica ^d	Arizona Fescue - Redondo	0.6	1.5
Oryzopsis hymenoides	Indian Ricegrass - Paloma	1.5	3.8
<u>FORBS</u>			
Aster glaucodes	Glaucous Aster	0.20	0.50
Heliomeris multiflora	Showy Goldeneye	0.12	0.30
Linum lewisii	Blue Flax - Appar	0.50	1.25
Penstemon palmeri	Palmer Penstemon	0.21	0.53
Solidago rigida	Rigid Goldenrod	0.17	0.43
<u>SHRUBS</u>			
Atriplex canescens	Fourwing Saltbush	<u>Planting Rate</u>	
Cercocarpus montanus	Birch-leaf Mountain Mahogany	<u>Stems / acre</u>	
Chrysothamnus nauseosus	Rubber Rabbitbrush	---200---	
Krascheninnikovia lanata	Winterfat	---400---	
Rhus trilobata	Three-leaf Sumac	---100---	

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded

^d More aggressive species that will be sown only in every third drill furrow.

Table E-6. Plant Material Mixes - Moist Shrubland

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	Seeding	
		Rate (PLS lb/ac)^b	B'cast^c
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Agropyron trachycaulum	Slender Wheatgrass - Premier, San Luis	0.3	0.8
Festuca rubra	Creeping Red Fescue - Dawson	0.1	0.3
Poa compressa	Canada Bluegrass - Reubens	0.02	0.05
		Planting Rate	
		<u>Stems / acre</u>	
Acer glabrum	Mountain Maple	---40---	
Amelanchier alnifolia	Serviceberry	---60---	
Crataegus erythropoda	Red Hawthorn	---40---	
Physocarpus monogynus	Ninebark	---40---	
Prunus americanus	American Plum	---60---	
Prunus virginiana	Black Chokecherry	---60---	
Quercus gambelii	Gambel's Oak	---60---	
Ribes aureum	Golden Currant	---40---	
Ribes inerme	Gooseberry	---40---	
Symphoricarpos albus	Common Snowberry	---40---	
Symphoricarpos occidentalis	Western Snowberry	---40---	

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded

Table E-7. Plant Material Mixes - Herbaceous Wetland

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	<u>Seeding Rate (PLS lb/ac)^b</u>	
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Asclepias incarnata	Swamp Milkweed	NA	0.25
Scirpus paludosus	Alkali Bulrush	NA	2.0
Typha latifolia	Broadleaf Cattail	NA	2.0
		<u>Planting Rate Stems / acre</u>	
Juncus arcticus	Baltic Rush	---100---	
Sagittaria latifolia	Common Arrowhead	---400---	
Scirpus acutus	Hardstem Bulrush	---500---	
Scirpus pungens	Threesquare	---500---	

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded

Table E-8. Plant Material Mixes - Wooded Wetland

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	<u>Seeding</u> <u>Rate (PLS lb/ac)^b</u>	
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Agropyron trachycaulum	Slender Wheatgrass - Premier, San Luis	0.3	1.0
Agrostis alba	Redtop	0.0003	0.001
Elymus canadensis	Canada Wildrye	0.3	1.0
		<u>Planting Rate</u> <u>Stems / acre</u>	
<u>SHRUBS</u>			
Alnus tenuifolia	Thinleaf Alder	---40---	
Betula fontinalis	Western Riverbirch	---40---	
Lonicera involucrata	Twinberry	---40---	
Pentaphylloides floribunda	Shrubby Cinquefoil	---20---	
Salix amygdaloides	Peachleaf Willow	---20---	
Salix exigua	Coyote Willow	---40---	
Salix interior	Sandbar Willow	---40---	
Salix bebbiana	Bebb Willow	---20---	
Swida sericea	Red-osier Dogwood	---20---	

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded

Table E-9. Plant Material Mixes -Cottonwood Grove

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	Seeding Rate (PLS lb/ac)^b	
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Agropyron trachycaulum	Slender Wheatgrass - Premier, San Luis	0.3	0.8
Elymus triticoides	Creeping Wildrye - Shoshoni	0.9	2.3
Festuca rubra	Creeping Red Fescue - Dawson	0.1	0.3
		Planting Rate	
<u>TREES</u>		<u>Stems / acre</u>	
Populus x acuminata	Lanceleaf Cottonwood ^d	---200----	
Populus deltoides	Plains Cottonwood ^d	---200----	

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded
^d 20% 3/4 inch caliper, 40% 1 to 1.5 inch caliper, and 40% 2 to 2.5 inch caliper.

Table E-10. Cover Crops for Use in Revegetation

<u>Crop</u>	<u>Date of Planting</u>	<u>Date of Seeding</u>	<u>Rate (PLS lb/ac)</u>
Wheat/Wheatgrass Hybrid			
("Regreen")	4/1 to 5/15	Next fall	35
	8/15 to 10/1	Next spring	35
Oats	4/1 to 5/15	Next fall	30
Winter Wheat	8/1 to 10/1	Next fall	25
Spring Barley	4/1 to 5/15	Next fall	30
Long-season (southern) sorghum	5/15 to 7/15	Next fall	10

Table E-11. Plant Material Mixes -Temporary Stabilization Sites

<u>Scientific Name</u>	<u>Common Name - Variety^a</u>	Seeding	
		Rate (PLS lb/ac)^b	
		<u>Drill</u>	<u>B'cast^c</u>
<u>GRASSES</u>			
Agropyron dasystachyum	Thickspike Wheatgrass - Critana	3.0	6.0
Agropyron riparium	Streambank Wheatgrass - Sodar	3.0	6.0
Agropyron smithii	Western Wheatgrass - Arriba	3.0	6.0
Agropyron trachycaulum	Slender Wheatgrass - Premier, San Luis	3.0	6.0
Festuca rubra	Creeping Red Fescue - Dawson	1.0	3.0

^a Variety unnamed native unless specified ^b PLS = Pure Live Seed ^c Broadcast seeded

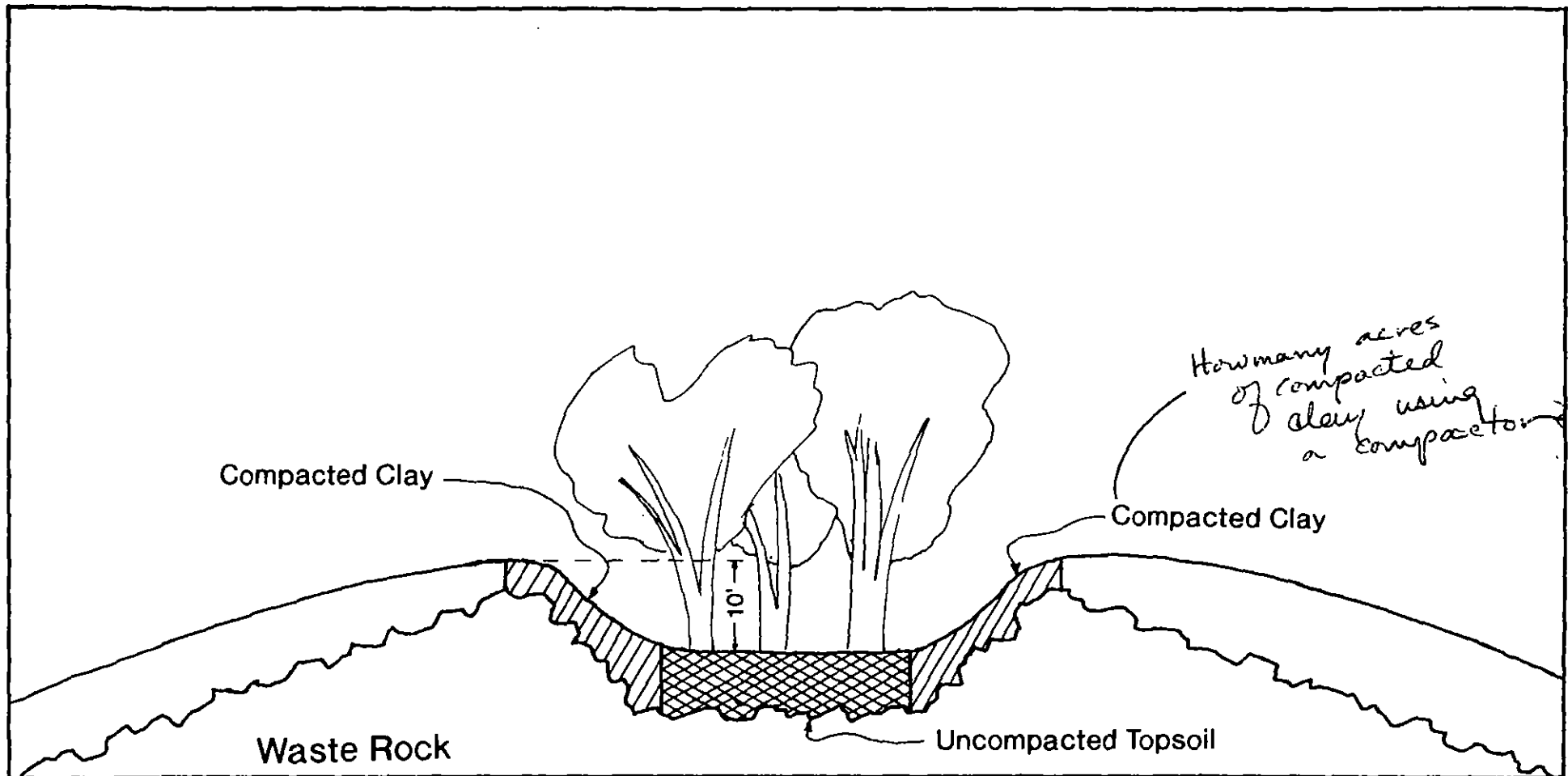


Figure E-1. Ridgetop Depressions for Cottonwood Grove Development

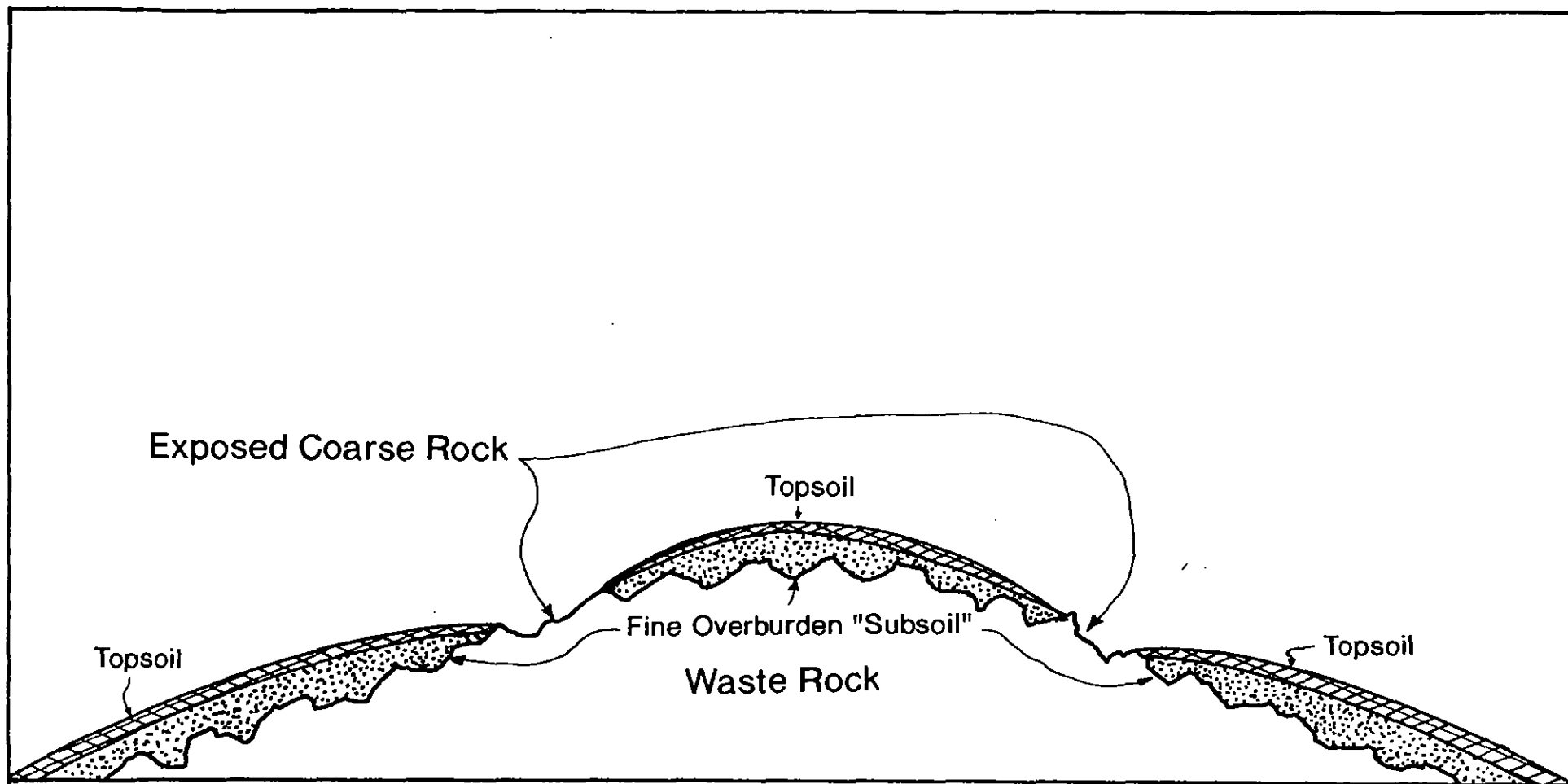


Figure E-2. Typical Cross-section of Constructed Topography with "Rimrock" Features

EXHIBIT G - WATER INFORMATION

- Water Augmentation permit from office of State Engineer
- Stormwater Discharge permit - proof of application

1.0 INTRODUCTION

The Dowe Flats Project is located in a gently sloping valley between Rabbit Mountain to the east and Indian Mountain to the west (Figure G-1). The valley slopes at an approximate grade of two percent or less to the south and is drained by intermittent streams of the St. Vrain Creek system. The unnamed Dowe Flats stream system flows southward through the property and intersects the Supply Ditch, an irrigation ditch located north of St. Vrain Creek along the southern edge of the property. The principle drainage in the vicinity of the project area is eastward flowing St. Vrain Creek. It originates in the mountains west of the town of Lyons and occurs approximately one mile to the south of the Cotenancy property.

The project area is used primarily for agriculture and livestock grazing. Boulder County Open Space land is situated to the northeast and east. Some private residences and the Syntex Chemicals (Syntex) waste facility is located to the north. The west side of Dowe Flats is comprised of undeveloped grazing land and a dairy farm. Other livestock properties are situated in the southern portion of the project area.

How far?
Locate on the map.
Any connection in ground or surface water w/ Dowe Flats?

The primary source of ground water in the project area is an unconsolidated shallow alluvial aquifer of Quaternary age that ranges between 12 and 30 feet thick. Cretaceous-age sedimentary formations, the Niobrara Formation and Pierre Shale, underlie the unconsolidated alluvium. Both formations are classified as aquifers with inadequate well yields of less than one gallon per minute (CGS, 1980). Domestic and stock wells are located on some of the properties especially north and south of the project area. In addition, there are approximately 157 monitoring wells permitted by Syntex in Section 4, Township (T) 3 North (N), Range (R) 90 West (W).

70 West?

A hydrologic monitoring network has been established to monitor the surface water and ground water at the Dowe Flats property. Monitoring locations (Figure G-2) were placed in upgradient and downgradient positions to establish baseline water quality.

2.0 SURFACE WATER

Site Description

The Dowe Flats Project area is located in a broad, gently-sloping drainage basin that is flanked on the north, east and west by more steeply-sloping terrain. The drainage system flows from ridges to the

north, east and west across grassy meadows on the alluvial and colluvial deposits that cover the central portion of Dowe Flats and continue south toward St. Vrain Creek, approximately one-half mile south of the project area. St. Vrain Creek is one of three major streams in Boulder County and is a tributary of the South Platte River. The St. Vrain Creek basin occupies almost the entire northern half of Boulder County with headwaters located along the Continental Divide approximately 23 miles west of Dowe Flats. Streamflow in the St. Vrain Creek basin varies seasonally, usually reaching maximum flow in June because of snowmelt runoff.

Dowe Flats is drained by an unnamed, intermittent stream system that experiences seasonal flow during the spring runoff period and essentially no flow during the remainder of the year. The drainages (Figure G-3) flow to the south toward St. Vrain Creek. Four irrigation ditches are present: Supply Ditch, the Highland Ditch, the Rough and Ready Ditch and the Palmerton Ditch. The northern and northwestern edges of the project area are bordered by the St. Vrain Supply Canal. The St. Vrain Supply Canal is a much larger water supply canal than the ditches located to the south and flows only in the summer. The Dowe Flats watershed is divided longitudinally into east and west subbasins by a north-south trending ridge (Figure G-3). This ridge also represents the western extent of the planned mining operations. The eastern subbasin is further divided into four subbasins based on the upstream and downstream locations of the proposed mining area.

Stream channels are predominantly grass covered with occasional sandy or rock-bottomed segments. The stream channels are straight to meandering, shallow and have a low gradient. The drainage pattern is braided in some areas. There are no incised channels in the planned mining area. Elevations along the drainages range from about 5,450 feet above mean sea level (ft msl) along the northern margin of the project area to about 5,300 ft msl at the southern margin.

Upstream of the project area, the stream channel is much narrower and has a steeper gradient as it gains in altitude toward its headwaters. Steeper gradients also are associated with the various headwater drainages on the steeper slopes of Rabbit and Indian Mountains, east and west of the relatively flat-lying central portion of Dowe Flats. The headwater of the major ephemeral stream in the eastern subbasin is located at an elevation of about 5800 ft msl on Indian Mountain, approximately one mile north of the project area. This stream flows by the Syntex site. The Dowe Flats watershed drains an area of approximately 4,346 acres (Table G-3) and is divided into five subbasins designated A, B₁, B₂, C, and D (Figure G-3).

Stream Flow Data

Because the drainages are all ephemeral and rarely flow, gaging stations have not been established on the project site.

100-Year Floodplain Limits

The 100-year floodplain of St. Vrain Creek does not extend onto the proposed mine area. No other 100-year floodplain has been identified on the property. Maps A, B, C and D show the 100-year flood plain of St. Vrain Creek in the vicinity of Dowe Flats.

Where are these?

Minimum Streamflow Standards

Because of the lack of perennial streams in the project area, minimum stream flow standards are not applicable.

Surface Water Quality

The surface water monitoring system installed for this project is comprised of three stations in the project area. An upgradient sampling station (DFSW-1) is located near the St. Vrain Supply Canal along the northern edge of the property. This station is situated in the ephemeral stream drainage that flows southward toward St. Vrain Creek. The two additional stations (DFSW-2 and DFSW-3) are located along the southern perimeter of the project area. They are situated downgradient from the ephemeral drainage that flows through the property. Water quality monitoring was begun on June 8, 1990 and has continued on a quarterly basis. A list of the parameters analyzed and the analytical method is summarized in Table G-1. The monitoring schedule to be implemented during the mine project is summarized in Table G-2.

are these located on the North side of St. Vrain Creek?

Historical Surface Water Quality

Historical surface water quality data for St. Vrain Creek are available in U.S. Geological Survey Open File Report 79-979. Two of the sample locations discussed in the report are applicable to the proposed mine site. Station SVC-1 is located upgradient of the proposed mine site on St. Vrain Creek near the town of Lyons, Colorado. The results of the water quality analysis from SVC-1 and SVC-2 are

Location of this station on Figure G-2?

summarized in Table G-4. SVC-2 is located on St. Vrain Creek at East County Line Road near Longmont.

Surface Water Quality Monitoring

What about 1991, 1992, 1993 sampling?

To date, three quarters of baseline surface water quality data have been obtained from existing surface water monitoring stations. The water quality data were collected quarterly, starting in June, 1990, pursuant to an application for a mine permit. Chemical analyses of the quarterly data provide background surface water quality data to provide baseline water quality. The chemical data are summarized in Table G-5.

Three surface water monitoring stations, designated DFSW-1, -2, and -3, are located (Figure G-2) in the vicinity of the mine. Station DFSW-1 is located upgradient of the proposed mine project and downgradient of the Syntex facility. DFSW-1 is situated about 100 feet south of DFMW-1 (NW 1/4, NE 1/4 Section 9, T3N, R70W in an ephemeral stream channel. Station DFSW-2 is located upstream of the proposed mine project, along the north bank of St. Vrain Creek, west of the bridge on the Southwestern Portland Cement entrance road (NE1/4, SW1/4, Section 21, T3M, R90W). Station DFSW-3 is located downstream of the mine project on the west bank of St. Vrain Creek north of the bridge on County Road 63 North (SE 1/4, NE 1/4, Section 27, T3N, R90W). The upgradient station provides background water quality data and stations DFSW-2 and DFSW-3 allow determination of the impact on St. Vrain Creek water quality from the proposed mining activities.

What about another location that is not a monitor well for Syntex? Could be screwy baseline data at some point.

Surface water quality data collected at stations DFSW-2 and DFSW-3 correlate moderately well and show a slightly bicarbonate water. Monitoring point DFSW-1 is located in an ephemeral stream and the analytical results do not correlate well with the other locations. The historical data listed in Table G-4 does however correlate moderately with data collected from DFSW-2 and DFSW-3. The Colorado Department of Health (CDH) Water Quality Commission has designated St. Vrain Creek as a Class 2 Recreational stream.

Surface Water Quality Compared to Drinking Water Standards

The CDH Water Quality Commission has established maximum contaminant levels (MCLs) and monitoring requirements to assure the safety of the public water supply. MCLs refer to several categories of chemicals, physical properties, radioactive, and microbiological parameters. Many of

these parameters have been analyzed during the quarterly monitoring conducted for the Dowe Flats mine permit application. The parameters, concentrations and MCLs are listed in Table G-6. Quarterly monitoring results indicate that the surface water at Dowe Flats meets all requirements for drinking water except for turbidity. The turbidity MCL was exceeded at each surface water station, except for station DFSW-1 during 3rd quarter, 1990, when the station was dry. A turbidity level of 5 turbidity units (TU) is acceptable under certain conditions. If these conditions could be met the turbidity level is exceeded only twice; once in the second quarter, 1990, at DFSW-2 and in the second quarter, 1990, at station DFSW-3.

which are?

Parameters not tested during quarterly monitoring include radioactivity, microbiological parameters, and a few specific organic compounds such as para-dichlorobenzene and total trihalomethanes.

analyze for these ASAP?

Historical Data Compared to Drinking Water Standards

The historical water quality data from St. Vrain Creek (Table G-4) does not readily compare to the drinking water standards. Many of the required analyses were not performed, and the analytical methods that were employed were generally not consistent with CDH monitoring requirements.

3.0 GROUND WATER

Hydrogeology

The project site is situated along the axis of the doubly-plunging Dowe Flats syncline in a valley mantled by unconsolidated valley fill deposits of Quaternary age. Beneath the valley fill deposits are folded beds of the Pierre and Niobrara formations of Cretaceous age. Older strata are exposed east and west of the project site.

The axis of the Dowe flats syncline trends approximately north-south through Dowe Flats. It plunges to the north along a line between sections 9 and 10, T3N, R70W and to the south in section 15, T3N, R70W. The syncline is asymmetrical with 10 to 20 degree dips along the west limb and vertical to overturned beds on the east. Interpretations of low altitude aerial photography suggest that there may be extremely tight folds superimposed on the eastern limb of the syncline. In addition, exploration drilling data indicates that the Niobrara Formation is tightly folded at depth along the eastern limb of

the Dowe Flats syncline. In similar geologic environments, tight folding such as this can result in fracturing along the fold axes.

North of the proposed site, the Dowe Flats syncline is truncated by a fault that crosses the southern parts of Sections 3 and 4, T3N, R70W. This fault is an easterly dipping thrust fault extending across the north end of Rabbit Mountain (Masters, 1957). A review of low altitude aerial photography revealed alignments which appear to support the presence of a fault across this area.

A summary of general water-bearing properties for the formations that outcrop in the proposed mine area is presented below followed by a detailed discussion of the Pierre Shale, Niobrara Formation and the unconsolidated valley fill.

Generalized geologic section for the site area and a description of the local aquifers are included as Figures G-4 and G-5 respectively. A summary of typical well yields for each unit is shown on these sections. The rock units in Dowe Flats were mapped by Southwestern Portland Cement Company in greater detail in 1985 and 1986. The formations and their thicknesses in Dowe Flats are presented in Table G-7. The thickness information was derived from outcrops, drill core, and cuttings.

Unconsolidated Valley Fill - The unconsolidated valley fill deposits are composed of alluvium, colluvium deposited in large part by sheet wash as well as by the ephemeral streams in Dowe Flats, and windblown material. These deposits range from about 12 to 30 feet thick where measured in boring and are poorly-sorted, crudely-bedded strata. The formation is thickest along the stream channels and in the southern part of Dowe Flats near the St. Vrain Creek floodplain. The valley fill becomes progressively thinner to the east and west along the steeper flanks of Dowe Flats.

Included in this formation is the weathered zone of bedrock found just below the valley fill. The well yields in the valley fill aquifer are reported by the Colorado Geological Survey (CGS) as being small to inadequate (less than 15 gallons per minute).

Floodplain Deposits - South of the mine property, St. Vrain Creek flows in a valley cut into Quaternary-age deposits of unconsolidated alluvium. These deposits are thicker than the valley fill deposits associated with the proposed mine area at Dowe Flats and are composed of stream deposited boulders, gravel, sand and some silt and clay. This type of deposit occurs in present and ancestral stream valleys, in terraces along present stream valleys and on the slopes of the foothills. In general,

floodplain deposits are considered by the CGS to be the principal water table aquifers in the Boulder - Greeley - Fort Collins area and have well yields of medium to large in the plains (15 gpm to greater than 100 gpm) and small to medium (less than 100 gpm) in the mountains.

Pierre Shale - The Pierre Shale of Cretaceous age outcrops or subcrops in the southern half of the Dowe Flats project area. Regionally, it is approximately 8,000 feet thick and is comprised of dark gray to black marine shale with some sandstone beds, the most prominent being the Hygiene Sandstone Member located about 1500 feet above the base of the formation. The formation yields very small amounts of water of poor quality to wells from weathered zones and the Hygiene Sandstone. The yields are reported by the CGS as being inadequate (less than one gallon per minute).

Niobrara Formation - The Niobrara Formation of Cretaceous age lies below the Pierre Shale and subcrops along the north, east and western flanks of the Dowe Flats syncline. The formation is composed of calcareous marine shale and limestone that yields very small quantities of water to wells. This formation is reported by the CGS to have inadequate well yields (less than 1 gpm). Figure G-6 is a geologic section showing the average thickness of the minable units within the Niobrara Formation.

Benton Formation - Also called the Benton Shale; this formation of Cretaceous age consists of marine shale, limestone, siltstone and a thin sandstone unit (the Codell Sandstone) near the top. It is reported to have inadequate well yields similar to the Niobrara Formation. This formation outcrops outside the eastern and western margins of the proposed mine area.

Dakota Group - Also known as the Dakota Sandstone; there are five members in this group consisting of sandstone, shale, conglomerate and silty carbonaceous shale. The Dakota yields small to moderate (less than 100 gpm) quantities of water to wells near the outcrop. This is the resistant rock which caps the hogback ridges along the western flank of the Dowe Flats valley and Rabbit Mountain along the eastern flank.

The unconsolidated valley fill material is the principal aquifer at Dowe Flats. Four monitoring wells were installed in these deposits to evaluate the local hydrology. The wells were completed in the valley fill material and the weathered bedrock zone just below it. Well completion details are included in "Report on Groundwater Monitor Well Installation, Dowe Flats Project" by Western Technologies Inc.

Results
of Wetlands
Re-evaluation?
Where are
wetlands that
exist now

Water level information has been obtained during each monitoring period. A contour map of the potentiometric surface has been constructed from the current and past water level data.

The Pierre Shale unconformably underlies the valley fill at Dowe Flats. Because it is predominantly comprised of shale, this formation is not considered water-bearing. Wells south of the project produce small quantities of water from weathered zones in the formation. Below the zone of surficial weathering the Pierre Shale is one of the major low permeability formations along the Front Range because of its high clay content.

The Niobrara Formation lies below the Pierre Shale and is exposed in the southern part of the project area and in the extreme northern part of the project area. There are also scattered outcrops of Niobrara along the higher ridges. This formation also has low permeability and is expected to produce water only from the weathered layer or from zones of secondary permeability such as solution cavities or fractures. Solution cavities are not known to exist at Dowe Flats. Fracture systems may be present, especially on the eastern side of the valley. The limestone pits south of Colorado Highway 66 are excavated in the Niobrara Formation. These pits yield little water from the limestone strata. Because of the thickness and the low permeability of the Niobrara Formation, it is anticipated that none of the deeper aquifers below the Niobrara will be impacted by the proposed quarry operation at Dowe Flats.

Recharge to the unconsolidated valley fill aquifer at Dowe Flats is principally from snowmelt and rainfall. Other less significant sources are streamflow and overlying aquifers. The USGS indicates in Water-Resources Investigation 83-4058 that water seeps from numerous reservoirs and irrigation canals into weathered bedrock. The St. Vrain Supply Canal, located on the northern margin of the site, is a potential source of limited amounts of seepage. This canal flows in the spring and summer and is dry during the winter months.

Springs and Seeps

The USGS topographic map of Dowe Flats identifies two springs in the Dowe Flats project area. The locations of these springs are shown on Figure G-3. Low altitude aerial photos (approximate scale: 1"=400') were examined for evidence of additional springs. No evidence was found. Because the streams are intermittent, there are areas where the emergence of the water in the stream bed exhibits a seep-like appearance. This can be observed during the spring and summer in the stream channel downstream from monitoring well DFMW-1. Standing water is present in the stream bed for a

distance of about 300 to 400 yards beginning at a point approximately 25 yards downstream from monitoring well DFMW-1. This area is mapped as a wetland (Exhibit J) and apparently is a localized area where the water table is higher than the elevation of the stream bed. This stretch of drainage is wet in the spring and summer and is dry during the winter.

One of the springs mapped by the USGS is located in a westerly flowing side drainage on the western side of Rabbit Mountain (NW1/4,NW1/4,NE1/4 Section 15, T3N, R70W). This spring occurs on land east of the Cotenancy property. The spring appears to issue from the contact of the Upper and Lower Muddy Members of the Dakota Formation. There is evidence on the aerial photographs that suggests a minor fault may coincide with the spring. A possible fault trace has been mapped on the aerial photos that trends across the spring location. The water in the spring appears to pond in a small area and flow a short distance downstream. Before reaching the Cotenancy property, the water apparently infiltrates into the unconsolidated alluvium and does not resurface on the Dowe Flats property.

The other mapped spring is located in the north-central portion of the Cotenancy property (NE1/4,SE1/4,SE1/4 Section 9, T3N, R70W). Examination of the aerial photographs indicates that the spring may be related to a contact between limestone and shale in the Niobrara Formation. The spring appears to be small and consists of a circular area that apparently has been scraped to provide a shallow basin for livestock use. This spring does not appear to contain any appreciable amounts of water during the winter months.

Ground Water Quality

The Dowe Flats project groundwater monitoring system is comprised of four ground water monitoring wells located up- and down-gradient of the proposed project area. One well is upgradient of the proposed quarry site and the remaining three monitoring wells are situated in downgradient positions. These wells are used to establish background water quality and piezometric conditions for the shallow alluvial aquifer. The locations of the four ground water monitoring wells are shown on Figure G-2. Monitoring and sampling activities of the surface and ground water were initiated on June 8, 1990 and have continued on a quarterly basis.

Baseline Data

Three sets of baseline ground water quality data have been obtained from four monitoring wells installed in the vicinity of the proposed mine project. The monitoring well locations are shown on Figure G-2. Monitoring well DFMW-1 is located in an ephemeral stream channel near the northern property boundary upgradient of the proposed site. DFMW-1 will provide background data for the shallow aquifer. Monitoring well DFMW-2 is located downgradient of the project area near the southwest corner of the site between Highway 66 and the Palmerton Ditch. Monitoring wells DFMW-3 and 4 are downgradient of the project area. DFMW-3 is located near the southeast corner of Garner Dairy north of the Rough and Ready Ditch. Monitoring well DFMW-4 is located near the southeast corner of the site along the south bank of the Palmerton Ditch. Data from the sampling events is summarized in Tables G-8 and G-9.

The water quality analyses for each sampling event were relatively consistent and indicate background conditions. No elevated concentrations of contaminants have been detected at concentrations above the EPA practical quantitation limit (PQL). Data indicate that the shallow alluvial aquifer is predominately bicarbonate.

Organic and metals analyses indicate no constituents at concentrations that would be considered above background. Major ions and other parameters are at concentrations that could be considered normal. The total suspended solids concentrations in two of the ground water monitoring wells were lower in the third quarter compared to the first quarters. This can be attributed to the continued development of the wells prior to sampling.

Major ion balance calculations were made to determine the completeness of the ground water analyses. Results of the ion balance calculations are presented in the quarterly monitoring reports. Groundwater samples are electrically neutral, so that total charges on cations and anions reported in the analyses should be equal. The total positive and negative charges are obtained by summing the equivalent concentrations of cations and anions. The ion balance error is normally expressed by the difference as a percentage of the sum and should be less than about 7.5 per cent error (Llyod and Heathcote, 1985). Ion balance for the third quarter analyses at Dowe Flats are within the acceptable range.

pH, temperature and specific conductance also were measured during each sampling event prior to sampling the water. Groundwater pH varies from 6.90 to 7.57. The specific conductance varies from

594 to 644 in DFMW-1, from 332 to 384 in DFMW-2, 1940 to 2252 in DFMW-3 and 643 to 765 in DFMW-4.

Ground Water/Surface Water Quality Comparison

Inorganic analyses of the surface and ground water samples indicate that the ground water samples contained elevated levels of calcium, and higher hardness concentrations than the surface water samples. Furthermore, magnesium, potassium, and sodium concentrations were higher at DFSW-1 than at DFSW-2 and DFSW-3. Trace levels of cadmium, iron, lead, manganese, mercury, and zinc were found in some monitoring well samples. Trace levels of iron, lead, manganese, mercury, and zinc were detected in the surface water samples.

Anion concentrations typically were higher in the ground water samples. Bicarbonate, chloride, and sulfate levels were significantly greater in the monitoring wells than at surface water monitoring locations. Concentrations of nitrates at all surface water and ground water monitoring locations were less than 0.15 ppm ($<0.15\text{ppm}$). Concentrations of nitrates ranged from 0.16 ppm at monitoring locations DFSW-2, DFSW-3 and DFMW-1, to <0.10 ppm at DFSW-1 and DFMW-3. pH concentrations for surface and ground water samples ranged from 6.99 to 7.54. Because of the pH concentrations, carbonates were not detected in any of the surface or ground water samples. Specific conductance, alkalinity, total suspended solids, total dissolved solids and turbidity values were higher in the ground water samples than in the surface water samples.

Ground Water Quality Monitoring Compared to Drinking Water Standards

The CDH Water Quality Commission has established MCLs and monitoring requirements to assess the safety of public water supply. The MCLs refer to several categories of chemicals, physical properties, radiation, and microbiological categories. Many of these parameters have been tested during the quarterly monitoring conducted for the mine permit at Dowe Flats. The physical properties and chemicals were analyzed and the MCLs are listed on Table G-9. The quarterly results indicate that the groundwater at Dowe Flats meet all requirements for drinking water except for turbidity. The turbidity levels in all of the wells exceed the MCL of 1TU. However, the turbidity requirement is reserved for surface water sources, and the turbidity measured in the wells could possibly be affected by monitoring well construction, well materials, or well development.

Berham is this a problem?

Parameters not tested during the quarterly monitoring include radioactivity, microbiological constituents, and a few specific organic compounds such as para-dichlorobenzene and total trihalomethanes.

Water Rights

Data obtained from the Colorado Division of Water Resources' files indicate numerous water rights on or in the vicinity of the Dowe Flats project. These rights are summarized in Table G-10.

A detailed description of the water rights location was not always stated in the files, so it is difficult to determine which of the rights are on or adjacent to the project. The following water rights however, are on or near the proposed project:

Lower Bullock Ditch,
Upper Bullock Ditch,
Outlet Ditch and PL.,
Longmont Well No. 4,
Martin Marietta Cem Lake,
Orth Seepage Ditch,
Weese Ditch,
Highland Ditch,
Stamp Drain Line 5,
Orth Pipeline, and
Orth Res. No.7.

The water rights listed in Table G-10 are primarily surface water. Ground water is accessed by the Longmont Wells. The Colorado Division of Water Resources data did not provide a listing of the aquifer accessed by these wells.

Figure G-10 presents the location of well permits and irrigation ditches within a one mile perimeter of the project site. With the exception of the monitoring wells on site, there are no other well permits within the boundary of the proposed mining operation. The majority of wells are located north and south of the site and are used for domestic purposes or as monitoring wells for the Syntex Chemicals site north of the project area. There are a total of 157 well permits in Section four where the Syntex site is located.

The wells with the best production are located along St. Vrain Creek and presumably are completed in the unconsolidated floodplain deposits.

USGS Open File Report 78-567 presents hydrologic data from water table aquifers in the area. Eight wells in the southern part of Dowe Flats, south of the proposed mining area, are included in the report. These data include the aquifer the well is completed in, depth of well and water, and chemical analyses. Three of the wells reportedly are completed in alluvial and terrace gravel, three in the Pierre Shale and two in dune sand. All wells are less than 81 feet in depth. The wells completed in the Pierre Shale are inferred to be completed in the weathered zone or in the Hygiene Sandstone member.

EXHIBIT G

LIST OF TABLES

- G-1 Water Quality Monitoring Parameters
- G-2 Monitoring Schedule
- G-3 Water Shed Acreage Calculation
- G-4 Historical Surface Water Quality
- G-5 Baseline Surface Water Quality
- G-6 Surface Water Quality Compared to Drinking Water Standards
- G-7 Stratigraphic Column for Dowe Flats Region
- G-8 Baseline Groundwater Quality
- G-9 Groundwater Quality Compared to Drinking Water Standards
- G-10 Existing Water Rights in the Vicinity of Dowe Flats

TABLE G - 1
WATER QUALITY PARAMETERS

<u>ORGANIC PARAMETERS (1)</u>	<u>INORGANIC PARAMETERS (2)</u>	<u>ANIONS (4)</u>
CHLOROMETHANE	ALUMINUM	CARBONATE
BROMOMETHANE	ARSENIC	BICARBONATE
VINYL CHLORIDE	BORON (3)	CHLORIDE
CHLOROETHANE	CADMIUM	NITRITE
METHYLENE CHLORIDE	CALCIUM	NITRATE
ACETONE	CHROMIUM (+6)	SULFATE
CARBON DISULFIDE	COPPER	
1,1-DICHLOROETHENE	HARDNESS (Ca)	
1,1-DICHLOROETHANE	IRON	<u>OTHER</u>
		<u>PARAMETERS (5)</u>
TRANS 1,2-DICHLOROETHENE	LEAD	pH (standard units)
CHLOROFORM	MAGNESIUM	
1,2-DICHLOROETHANE	MANGANESE	
2-BUTANONE	MERCURY	SPECIFIC
		CONDUCTANCE
1,1,1-TRICHLOROETHANE	NICKEL	(uMHO/cm@25C)
CARBON TETRACHLORIDE	POTASSIUM	ALKALINITY
BROMODICHLOROMETHANE	SODIUM	(mg CaCO ₃ /L)
VINYL ACETATE	ZINC	TOTAL DISSOLVED
		SOLIDS
1,2-DICHLOROPROPANE		(mg/L)
TRANS 1,3-DICHLOROPROPENE		TOTAL
		SUPENDED
		SOLIDS
TRICHLOROETHENE		(mg/L)
1,1,2, TRICHLOROETHANE		TURBIDITY
		(NTU:HL)
BENZENE	<u>METHODS</u>	
DIBROMOCHLOROETHANE	(1) EPA 8260(8240)	
CIS, 1,3-DICHLOROPROPENE	(2) EPA 600/4-79-20	
2-CHLOROETHYL VINYL ETHER	(3) Standard Method 404A	
BROMOFORM	(4) Bicarbonate - Standard Methods 403	
4-METHYL - 2-PENTANONE	Chloride - EPA 300.0	
2-HEXANONE	Nitrate - EPA 300.0	
1,1,2,2,-TETRACHLOROETHANE	Nitrite - EPA 300.0	
TETRACHLOROETHENE	Sulfate - EPA 300.0	
TOLUENE	(5) pH - EPA 150.1	
CHLOROBENZENE	Specific Conductance(uMHO/cm@25C)EPA	
ETHYL BENZENE	9050	
STYRENE	Alkalinity(mg CaCO ₃ /L)Standard Method 403	
TOTAL XYLENES	Total Dissolved Solids Standard Method 209B	
TETRAHYDROFURAN	Total Suspended Solids Standard Method 209C	
DIETHYL ETHER	Turbidity Standard 2130	

TABLE G-2

PROPOSED MONITORING AND SAMPLING SCHEDULE

<u>TYPE OF MONITORING</u>	<u>FREQUENCY</u>	<u>MONITORING/SAMPLING EQUIPMENT</u>
Surface Water Quality	Quarterly	Grab Sample, Sample Jars
Groundwater Quality	Quarterly	Bailers, Sample Jars
Groundwater Levels	Monthly	Water Level Probe

TABLE G-3
SUBBASIN AREA CALCULATIONS

<u>SUBBASIN</u>	<u>CALCULATED AREA (ACRES)*</u>
A	410
B1	394
B2	1,652
C	627
D	1,263
TOTAL	4,346

* Areas calculated with planimeter

TABLE G-4
HISTORICAL WATER QUALITY ANALYSIS
MARCH 10, 1975
(USGS 79-979)

SVC-1

<u>PARAMETER</u>	<u>CONCENTRATION</u> (mg/l) at 15° C
Silica (Dissolved)	4.2 SiO ₂
Calcium (Dissolved)	12 Ca
Magnesium (Dissolved)	2.0 Mg
Sodium (Dissolved)	2.4 Na
Potassium (Dissolved)	1.0K
Bicarbonate	44 HCO ₃
Alkalinity	36 CaCO ₃
Sulfate (Dissolved)	6.3 SO ₄
Chloride (Dissolved)	.9 Cl
Fluoride (Dissolved)	.1 F
Nitrogen (Dissolved)	.01N
Orthophosphorous (Dissolved)	.00 P
Solids Constituents (Dissolved)	51
Hardness	38 CaCO ₃
Hardness (noncarbonated)	2 CaCO ₃
Sodium Adsorption Ratio	.2
Specific Conductance	90 micromhos

TABLE G-5

(continued)

BASELINE SURFACE WATER QUALITY MONITORING
1990
WITH SAMPLING EVENT COMPARISON

ORGANIC PARAMETERS (1)	DFSW-1			DFSW-2			DFSW-3		
	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90
	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)
CHLOROMETHANE	U	BJ	*	U	U	U	U	U	U
BROMOMETHANE	U	U	*	U	U	U	U	U	U
VINYL CHLORIDE	U	U	*	U	U	U	U	U	U
CHLOROETHANE	U	U	*	U	U	U	U	U	U
METHYLENE CHLORIDE	U	U	*	U	2J	2J	U	U	U
ACETONE	U	U	*	U	U	U	U	U	U
CARBON DISULFIDE	U	U	*	U	U	U	U	U	U
1, 1-DICHLOROETHENE	U	2J	*	U	U	U	U	U	U
1, 1-DICHLOROETHANE	U	U	*	U	U	U	U	U	U
TRANS 1,2-DICHLOROETHENE	U	U	*	U	U	U	U	U	U
CHLOROFORM	U	U	*	U	U	U	U	U	U
1,2-DICHLOROETHANE	U	U	*	U	U	U	U	U	U
2-BUTANONE	U	U	*	U	U	U	U	U	U
1,1,1-TRICHLOROETHANE	U	U	*	U	U	U	U	U	U
CARBON TETRACHLORIDE	U	U	*	U	U	U	U	U	U
BROMODICHLOROMETHANE	U	U	*	U	U	U	U	U	U
VINYL ACETATE	U	U	*	U	U	U	U	U	U
1,2-DICHLOROPROPANE	U	U	*	U	U	U	U	U	U
TRANS 1,3-DICHLOROPROPENE	U	U	*	U	U	U	U	U	U
TRICHLOROETHENE	U	1J	*	U	U	U	U	U	U
1,1,2-TRICHLOROETHANE	U	U	*	U	U	U	U	U	U
BENZENE	U	1J	*	U	U	U	U	U	U
DIBROMOCHLOROMETHANE	U	U	*	U	U	U	U	U	U
CIS,1,3-DICHLOROPROPENE	U	U	*	U	U	U	U	U	U
2-CHLOROETHYL VINYL ETHER	U	U	*	U	U	U	U	U	U
BROMOFORM	U	U	*	U	U	U	U	U	U
4-METHYL - 2-PENTANONE	U	U	*	U	U	U	U	U	U
2-HEXANONE	U	U	*	U	U	U	U	U	U
1,1,2,2-TETRACHLOROETHANE	U	U	*	U	U	U	U	U	U
TETRACHLOROETHENE	U	U	*	U	U	U	U	U	U
TOLUENE	U	1J	*	U	U	1BJ	U	U	U
CHLOROBENZENE	U	U	*	U	U	U	U	U	U
ETHYL BENZENE	U	U	*	U	U	U	U	U	U
STYRENE	U	U	*	U	U	U	U	U	U
TOTAL XYLENES	U	U	*	U	U	U	U	U	U
TETRAHYDROFURAN	U	U	*	U	U	U	U	U	U
DIETHYL ETHER	U	U	*	U	U	U	U	U	U

NOTES:

U = Compound analyzed for, but not detected above reporting limits.

J = Indicates an estimated value when the compound is detected, but is below the EPA Practical Quantification Limit (PQL).

BJ = Compound found in blank, however, concentration is below the EPA Practical Quantification Limit (PQL).

* = Not analyzed.

EAI = Evergreen Analytical Inc.

AI = Analytica, Inc.

(1) = EPA METHOD 8260 (8240)

TABLE G-5

**BASELINE SURFACE WATER QUALITY MONITORING
1990
WITH SAMPLING EVENT COMPARISON**

INORGANIC PARAMETERS (2)	DFSW-1			DFSW-2			DFSW-3		
	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)
ALUMINUM	U	U	*	U	U	U	U	U	U
ARSENIC	U	U	*	U	U	U	U	U	U
BORON (3)	1.2	0.27	*	1.1	U	U	1.0	U	U
CADMIUM	U	U	*	U	U	U	U	U	U
CALCIUM	81.0	97.0	*	5.0	12.0	24.0	4.0	15.0	44.0
CHROMIUM	U	U	*	U	U	U	U	U	U
CHROMIUM(+6)	*	U	*	*	U	U	*	U	U
COPPER	U	U	*	U	U	U	U	0.06	U
HARDNESS(Ca)	200.0	340.0	*	13.0	38.0	80.0	11.0	47.0	150.0
IRON	0.06	U	*	0.2	0.06	0.06	0.2	U	U
LEAD	U	U	*	0.007	0.009	0.009	U	0.005	U
MAGNESIUM	25.0	23.0	*	0.1	2.0	5.0	1.0	2.0	9.0
MANGANESE	0.1	0.2	*	U	U	U	U	U	U
MERCURY	U	U	*	0.001	U	U	U	U	U
NICKEL	U	U	*	U	U	U	U	U	U
POTASSIUM	1.1	1.0	*	0.95	0.90	0.9	2.6	1.0	0.9
SODIUM	15.0	13.0	*	3.0	2.0	6.0	2.0	3.0	10.0
ZINC	U	U	*	0.03	U	U	0.03	0.06	U

ANIONS (4)	DFSW-1			DFSW-2			DFSW-3		
	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)
CARBONATE	**	**	*	**	**	**	**	**	**
BICARBONATE	276	275	*	35.4	45.2	67.2	33.7	52.2	106.0
CHLORIDE	294	22.4	*	8.10	4.45	4.48	8.10	4.13	2.48
NITRITE	U	U	*	U	U	U	U	U	U
NITRATE	U	U	*	0.16	U	0.324	0.16	U	0.203
SULFATE	223	99.5	*	7.04	8.63	33.90	7.04	11.30	83.00

OTHER PARAMETERS (5)	DFSW-1			DFSW-2			DFSW-3		
	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90
pH (standard units)	7.45	7.62	*	7.07	7.87	7.47	7.10	7.80	7.63
SPECIFIC CONDUCTANCE (uMHO/cm@25C)	579.0	532.0	*	25.7	63.6	150.0	28.7	84.9	268.0
ALKALINITY (mg CaCO3/L)	226.0	225.0	*	29.0	37.0	55.0	27.6	42.8	86.9
TOTAL DISSOLVED SOLIDS (mg/L)	502.0	442.0	*	52.0	67.0	171.0	63.0	83.0	231.0
TOTAL SUSPENDED SOLIDS (mg/L)	30.7	8.5	*	5.3	5.5	8.5	4.7	U	U
TURBIDITY (NTU; HLT)	3.0	4.0	*	3.0	8.0	1.2	3.0	11.0	1.8

METHODS

(1) EPA 8260(8240)

(2) EPA 600/4-79-20

(3) Standard Method 404A

(4) Bicarbonate - Standard Method

Chloride - EPA 300.0

Nitrate - EPA 300.0

Nitrite - EPA 300.0

Sulfate - EPA 300.0

(5) pH - EPA 150.1

Specific Conductance (uMHO/cm @ 25C) - EPA 90

Alkalinity (mg CaCO3/L) - Standard Method 403

Total Dissolved Solids - Standard Method 209B

Total Suspended Solids - Standard Method 209C

TABLE G-6

SURFACE WATER QUALITY COMPARED TO DRINKING WATER STANDARDS

<u>Analyzed Parameter</u>	<u>Drinking Water Standards Concentration</u>	<u>Dowe Flats Water Quality Concentration¹</u>		
turbidity	ITU(5)	4.0	8.0	11.0
arsenic	.05	U ⁶	U	U
cadmium	.010	U	U	U
Chromium	.05	U	U	U
lead	.05	U	0.009	0.005
nitrate as N	10.0-20.0	U	0.324	0.203
mercury	.002	U	0.001	U
sodium	M ²	15.0	6.0	10.0
Benzene	.005	1J	U	U
Carbon Tetrachloride	.005	U	U	U
1,1-Dichloroethene	.007	2J	U	U
1,2-Dichloroethane	.005	U	U	U
1,1,1-Trichloroethane	0.200	U	U	U
Trichloroethene	0.005	1J	U	U
Vinyl chloride	0.002	U	U	U
Toluene	LUR ³	1J	U	U
Bromodichloromethane	TTHM ⁴	U	U	U
TTHM	0.10			
hardness	M			
pH	M	7.45	7.07	7.10
		7.62	7.87	7.80
calcium	M	81.0	5.0	4.0
		97.0	24.0	44.0
alkalinity	M	225.0	29.0	27.6
temperature	M	226.0	55.0	86.9

1. Concentrations are highest values recorded or highest and lowest for that station over three quarters of monitoring.
2. M: monitoring test and submit analysis
3. LUR: listed unregulated report detection
4. TTHM: trihalomethanes sum of bromodichloromethane, dibromochloroemethane, dibromomethane (bromoform) and trichloromethane
5. TU: turbidity unit
6. U: not detected

Parameters Not Analyzed coliform, Langelier Index (LI), and Radioactivity

TABLE G-7

STATIGRAPHIC COLUMN FOR DOWE FLATS

<u>Formation</u>	<u>Average Thickness (ft)</u>
QUATERNARY VALLEY FILL DEPOSITS	0-50
PIERRE FORMATION	8,000
NIOBRARA FORMATION	
SMOKEY HILL MEMBER	
FOURTH RIDGE	12-40
3-4 INTER-RIDGE	55
THIRD RIDGE	30-34
2-3 Inter Ridge	100
SECOND RIDGE	10
1-2 Inter-Ridge	100
FT. HAYS MEMBER	
FIRST RIDGE (Hi-Cal)	15
BENTON FORMATION	
CODELL SANDSTONE	24
CARLILE SHALE	26
GREENHORN LIMESTONE	220
GRANEROS SHALE	165
MOWRY SILTSTONE	22
DAKOTA GROUP	
UPPER MUDDY SANDSTONE	46
LOWER MUDDY SANDSTONE	94
SKULL CREEK SHALE	95
PLAINVIEW FORMATION	80
LYTLE SANDSTONE	49

TABLE G-8

**BASELINE GROUNDWATER QUALITY MONITORING
1990
WITH SAMPLING EVENT COMPARISON**

	DFMW-1			DFMW-2			DFMW-3			DFMW-4		
INORGANIC PARAMETERS (2)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)
ALUMINUM	U	U	U	U	U	U	U	U	U	U	U	U
ARSENIC	U	U	U	U	U	U	U	U	U	U	U	U
BORON (3)	0.9	0.26	U	1.1	0.13	U	1.0	0.30	U	0.9	0.22	U
CADMIUM	0.006	U	0.009	U	U	U	U	U	U	U	U	U
CALCIUM	94.0	140.0	110.0	58.0	69.0	78.0	560.0	470.0	470.0	100.0	140.0	120.0
CHROMIUM	U	U	U	U	U	U	U	U	U	U	U	U
CHROMIUM(+6)	*	U	U	*	U	U	*	U	U	*	U	U
COPPER	U	U	U	U	U	U	U	0.03	U	U	U	U
HARDNESS(Ca)	240.0	470.0	390.0	150.0	220.0	260.0	1400	1600	1300.0	250.0	470.0	420.0
IRON	U	U	U	U	0.09	U	0.07	U	U	U	0.09	0.06
LEAD	0.005	0.008	U	0.01	U	U	0.005	U	U	0.007	U	U
MAGNESIUM	28.0	31.0	26.0	15.0	10.0	14.0	110.0	110.0	17.0	30.0	27.0	29.0
MANGANESE	U	U	U	U	U	U	0.05	0.06	U	0.10	0.3	0.2
MERCURY	U	U	U	U	U	U	U	0.003	U	U	0.0005	U
NICKEL	U	U	U	U	U	U	U	U	U	U	U	U
POTASSIUM	5.4	2.0	1.0	2.0	1.0	0.9	1.6	1.0	0.9	2.1	0.7	0.4
SODIUM	27.0	29.0	17.0	13.0	5.0	10.0	96.0	110.0	110.0	27.0	27.0	25.0
ZINC	U	U	0.05	U	U	0.03	0.04	0.03	0.03	0.07	U	U

	DFMW-1			DFMW-2			DFMW-3			DFMW-4		
ANIONS (4)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)	Jun-90 (mg/L)	Sep-90 (mg/L)	Dec-90 (mg/L)
CARBONATE	**	**	**	**	**	**	**	**	**	**	**	**
BICARBONATE	1917	291	253	365	230	226	439	306	293	7441	276	279
CHLORIDE	128.0	67.6	25.7	11.0	3.47	5.08	279.0	21.1	18.9	8.7	8.27	5.39
NITRITE	U	U	U	U	U	U	U	U	U	U	U	U
NITRATE	0.16	0.38	0.43	0.1	U	U	U	2.33	5.1	0.12	U	U
SULFATE	122.0	208.0	168.0	68.1	24.2	91.7	2020	1500	1300	349	298	264

TABLE G-8
(continued)

BASELINE GROUNDWATER QUALITY MONITORING
1990
WITH SAMPLING EVENT COMPARISON

OTHER PARAMETERS (5)	DFMW-1			DFMW-2			DFMW-3			DFMW-4		
	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90
pH (standard units)	7.35	7.20	7.31	7.54	7.56	7.57	7.00	6.90	7.05	7.40	7.30	7.45
SPECIFIC CONDUCTANCE (uMHO/cm@25C)	644	843	594	378	332	384	2130	2252	1940	673	765	643
ALKALINITY (mg CaCO ₃ /L)	1571	238	208	299	188	185	360	251	240	6100	228	228
TOTAL DISSOLVED SOLIDS (mg/L)	560	760	524	306	212	340	2702	2560	2590	514	652	658
TOTAL SUSPENDED SOLIDS (mg/L)	2596	130	49	627	406	404	1500	584	511	20723	5755	4660
TURBIDITY (NTU; HLU)	1500	200	60	11	190	210	980	12	150	9000	2400	4500

METHODS

- (1) EPA 8260 (8240)
- (2) EPA 800/4-79-20
- (3) Standard Method 404A
- (4) Bicarbonate - Standard Methods 403
 - Chloride - EPA 300.0
 - Nitrate - EPA 300.0
 - Nitrite - EPA 300.0
 - Sulfate - EPA 300.0
- (5) pH - EPA 150.1
 - Specific Conductance (uMHO/cm @ 25C) - EPA 9050
 - Alkalinity (mg CaCO₃/L) - Standard Method 403
 - Total Dissolved Solids - Standard Method 209B
 - Total Suspended Solids - Standard Method 209C
 - Turbidity - Standard Method 2130

TABLE G-8
(continued)

BASELINE GROUNDWATER QUALITY MONITORING
1990
WITH SAMPLING EVENT COMPARISON

ORGANIC PARAMETERS (1)	DFMW-1					DFMW-2			DFMW-3			DFMW-4		
	Jun-90	Aug-90	Aug-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90	Jun-90	Sep-90	Dec-90
	(ug/L)	EAI (ug/L)	AI (ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)	(ug/L)
CHLOROMETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
BROMOMETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
VINYL CHLORIDE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
CHLOROETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
METHYLENE CHLORIDE	U	U	8	U	U	U	U	U	U	U	U	U	U	U
ACETONE	18J	U	U	U	U	U	U	U	11J	U	U	U	U	U
CARBON DISULFIDE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1, 1-DICHLOROETHENE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1, 1-DICHLOROETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
TRANS 1,2-DICHLOROETHENE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
CHLOROFORM	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1,2-DICHLOROETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
2-BUTANONE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1,1,1-TRICHLOROETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
CARBON TETRACHLORIDE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
BROMODICHLOROMETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
VINYL ACETATE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1,2-DICHLOROPROPANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
TRANS 1,3-DICHLOROPROPENE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
TRICHLOROETHENE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1,1,2-TRICHLOROETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
BENZENE	1J	U	U	U	U	U	U	U	U	U	U	U	U	U
DIBROMOCHLOROMETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
CIS,1,3-DICHLOROPROPENE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
2-CHLOROETHYL VINYL ETHER	U	U	U	U	U	U	U	U	U	U	U	U	U	U
BROMOFORM	U	U	U	U	U	U	U	U	U	U	U	U	U	U
4-METHYL - 2-PENTANONE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
2-HEXANONE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
1,1,2,2-TETRACHLOROETHANE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
TETRACHLOROETHENE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
TOLUENE	18J	U	U	U	U	U	U	28J	U	U	U	18J	U	U
CHLOROBENZENE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
ETHYL BENZENE	1J	U	U	U	U	U	U	U	U	U	U	U	U	U
STYRENE	U	U	U	U	U	U	U	U	U	U	U	U	U	U
TOTAL XYLENES	5J	U	U	U	U	U	U	U	U	U	U	U	U	U
TETRAHYDROFURAN	U	U	U	U	U	U	U	U	U	U	U	U	U	U
DIETHYL ETHER	U	U	U	U	U	U	U	U	U	U	U	U	U	U

NOTES:

U = Compound analyzed for, but not detected above reporting limits.

J = Indicates an estimated value when the compound is detected, but is below the EPA Practical Quantification Limit (PQL).

BJ = Compound found in blank, however, concentration is below the EPA Practical Quantification Limit (PQL).

* = Not analyzed.

EAI = Evergreen Analytical Inc.

AI = Analytica, Inc.

(1) = EPA METHOD 8260 (8240)

TABLE G-9

GROUND WATER QUALITY COMPARED TO DRINKING WATER STANDARDS

<u>Analyzed Parameter</u>	<u>Drinking Water Standards Concentration</u>	<u>Dowe Flats Water Quality Concentration¹</u>			
turbidity	ITU(5)	1500,	210,	980	9000NTU ⁶
arsenic	.05	U ⁷	U	U	U
cadmium	.010	0.009	U	U	U
Chromium	.05	U	U	U	U
lead	.05	0.008	0.01	0.005	U
nitrate as N	10.0-20.0	0.43	0.1	5.1	0.12
mercury	.002	U	U	0.003	0.0005
sodium	M ⁴	29.0	13.0	110	27.0
Benzene	.005	1J	U	U	U
Carbon Tetrachloride	.005	U	U	U	U
1,1-Dichloroethene	.007	U	U	U	U
1,2-Dichloroethane	.005	U	U	U	U
1,1,1-Trichloroethane	0.200	U	U	U	U
Trichloroethene	0.005	U	U	U	U
Vinyl chloride	0.002	U	U	U	U
Toluene	LUR ²	1BJ ⁶	2BJ	U	1BJ
TTHM ³	0.10				
hardness	M				
pH	M	7.20	7.54	6.90	7.30
		7.35	7.57	7.05	7.45
calcium	M	94.0	58.0	470.0	100.0
		140.0	78.0	560.0	140.0
alkalinity	M	208	185	240	226
		1541	299	360	6100
temperature	M				

1. Concentration: Height concentration recorded per quarter of highest and lowest concentration recorded.
2. LUR: listed unregulated report detection
3. TTHM: trihalomethanes sum of bromodichloromethane, dibromochloromethane, dibromomethane (bromoform) and trichloromethane
4. M: monitoring test and submit analysis
5. TU: turbidity unit
6. BJ: Analyte detected in blank
7. U: not detected

Parameters Not Analyzed: coliform, Langelier Index (LI), and Radioactivity

TABLE G-10

EXISTING WATER RIGHTS IN THE VICINITY OF DOWE FLATS

<u>Name of Structure</u>	<u>Location</u> <u>TWN Rng Sec</u>	<u>Structure</u> <u>Type</u>	<u>Amount</u>
Lower Bullock Ditch	3N 70W 21	D	3.4000c
Upper Bullock Ditch	3N 70W 21	D	3.0000c
Pleasant V. Dr & Sep No.1	3N 70W 22	D	1.0000c
Pleasant V. Dr & Sep No.2	3N 70W 23	D	1.5000c
Pleasant V. Dr & Sep No.4	3N 70W 24	D	2.0000c
Pleasant Valley Seep No. 1	3N 70W 22	D	0.5000c
Pleasant Valley Seep No.2	3N 70W 23	D	1.0000c
Pleasant Valley Seep No.3	3N 70W 23	D	1.0000c
Pleasant V. Dr & Sep No.3	3N 70W 24	D	1.0000c
Pleasant V. Dr. No.1	3N 70W 17	D	0.5000c
Stamp Drainage Line #7	3N 70W 22	O	0.6200c
James Ditch	3N 70W 22	D	0.0044c
Outlet Ditch & PL	3N 70W 21	D	3.0000c
Upper Bullock Ditch	3N 70W 21	D	3.0000c
Longmont Drain	3N 70W 20	L	0.8900c
Longmont Well 1	3N 70W 20	W	2.2280c
Longmont Well 2	3N 70W 20	W	2.2280c
Longmont Well 3	3N 70W 20	W	2.2280c
Longmont Well 4	3N 70W 20	W	1.3380c
Longmont Well 5	3N 70W 20	W	0.8900c
Longmont Well 4	3N 70W 21	W	1.3369c
Martin Marietta C & M Lake	3N 70W 21	R	36.0000a
Rock'n WP Ranch Lake #2	3N 70W 22	R	550.0000a
Orth Seepage Ditch	3N 70W 21	E	32.6000c
Highland Overflow	3N 70W 20	O	5.0000c
Chapman McCaslin	3N 70W 22	D	98.1300c
Supply Ditch (523)	3N 70W 20	D	*
Highland Ditch (526)	3N 70W 20	D	*
Rough and Ready Ditch	3N 70W 20	D	*
St. Vrain Palmerton Ditch	3N 70W 20	D	*
Smead Ditch	3N 70W 20	D	16.2700c
St. Vrain Creek	3N 70W 22	O	0.0720c
Baker Wease D	3N 70W 22	D	2.80000c
Weese Private	3N 70W 21	D	3.9600c

TABLE G-10
(continued)

<u>Name of Structure</u>	<u>Location</u> <u>TWN Rng Sec</u>	<u>Structure</u> <u>Type</u>	<u>Amount</u>
Swede Ditch	3N 70W 20	D	0.2070c
Swede Ditch	3N 70W 20	D	15.4000c
Highland Ditch	3N 70W 21	D	23.570c
Stamp Drain Line 5	3N 70W 21	L	2.0000c
Stamp Drain Line 1	3N 70W 22	L	2.0000c
Stamp Drain Line 2	3N 70W 22	L	2.0000c
Stamp Drain Line 3	3N 70W 22	L	2.0000c
Stamp Drain Line 4	3N 70W 22	L	2.0000c
Stamp Res 1	3N 70W 22	R	0.2500a
Stamp Res 2	3N 70W 22	R	2.0000a
Stamp Res 3	3N 70W 22	R	2.0000a
Frontier Lyons Res	3N 70W 20	R	1,700.0000a
ORTH Pipeline	3N 70W 21	L	1.4000c
ORTH Res No.7	3N 70W 21	R	2.2440a

NOTE:

D - ditch

O - other

L - pipeline

W - well

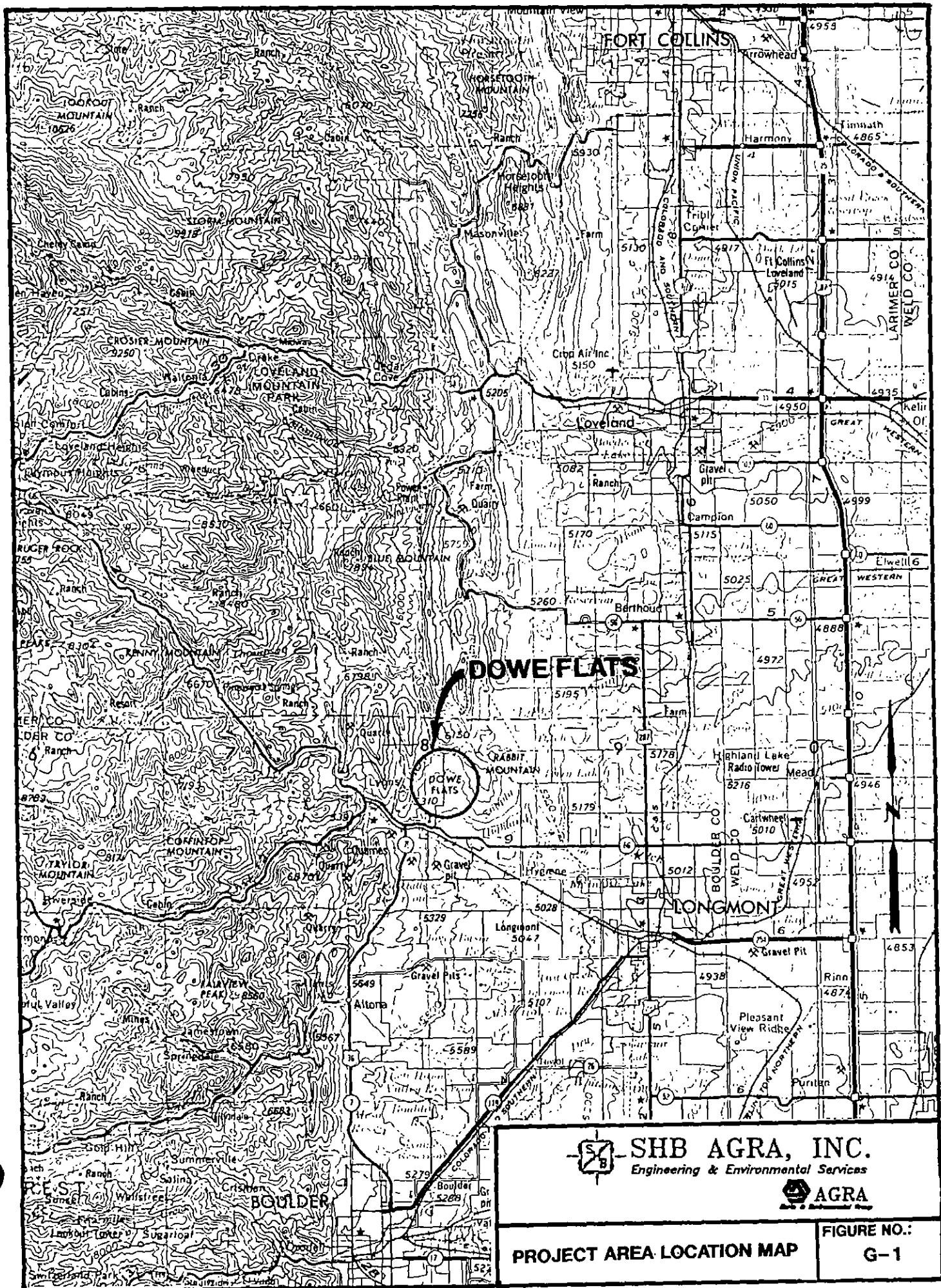
R - reservoir

E - seep

* - no amount listed

a - acre-feet per year

c - cubic feet per second

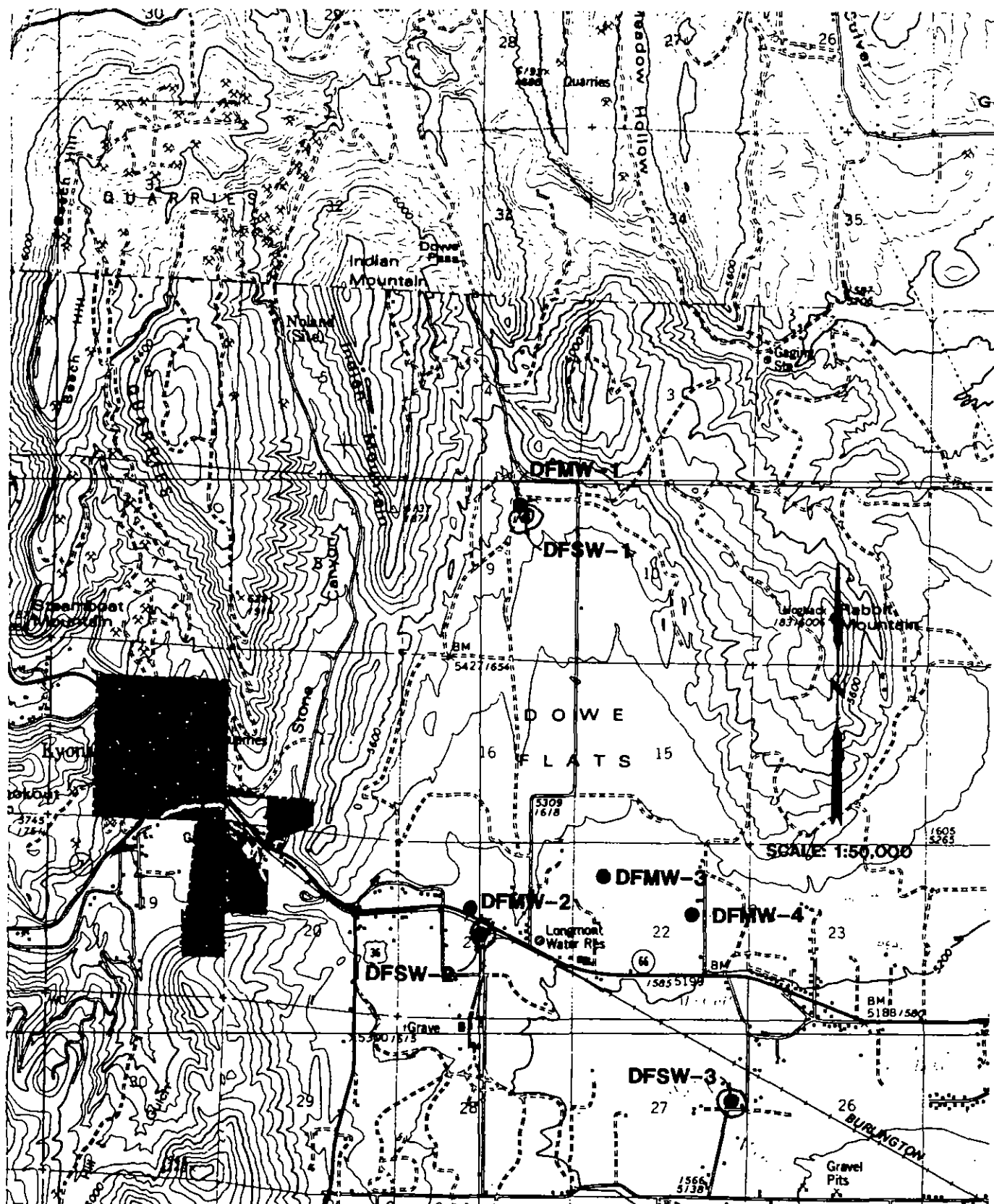


SHB AGRA, INC.
Engineering & Environmental Services



PROJECT AREA LOCATION MAP

FIGURE NO.:
G-1



- — DFMW-1 MONITORING WELL LOCATION
- — DFSW-3 SURFACE WATER MONITORING LOCATION

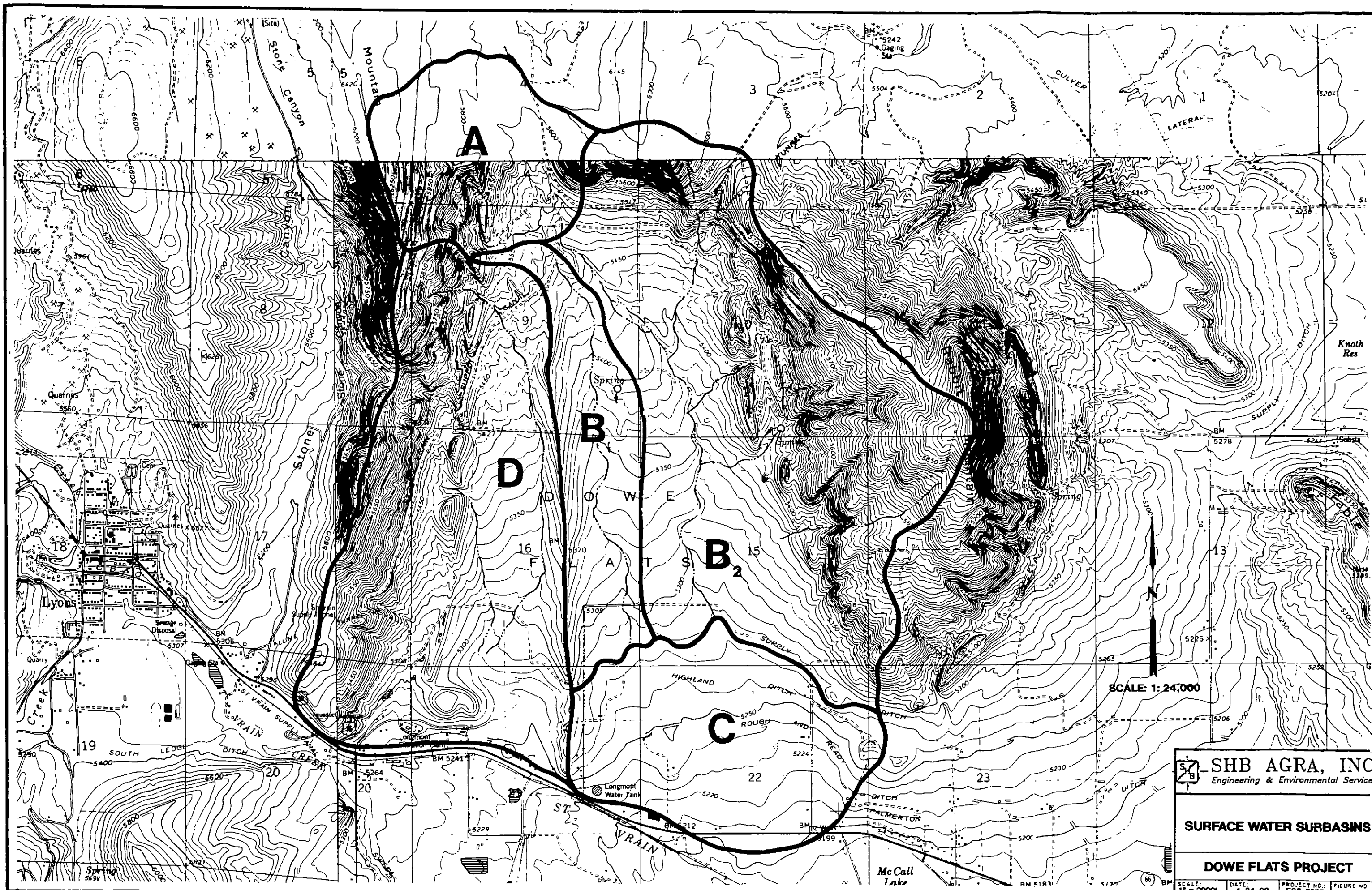
SOURCE: WESTERN TECHNOLOGIES INC.

SHB AGRA, INC.
Engineering & Environmental Services

**HYDROLOGIC MONITORING
LOCATIONS**

DOWE FLATS PROJECT

SCALE: 1"=1000' DATE: 1-21-93 PROJECT NO.: C92-7075 FIGURE NO. G-2



GENERALIZED SECTION OF GEOLOGIC FORMATIONS NEAR DOWE FLATS

Era	System or period	Series	Formation	Member	Typical thickness (feet)	Physical character	Water supply
Cenozoic	Quaternary	Recent	Alluvium (Qal)		25	Boulders, gravel, sand, silt, and clay	Yields moderate quantities of water to wells, springs, and seeps
		Pleistocene	Eolian silt and sand (Qes)		10	Poorly sorted sand and silt; contains some pebbles	Generally lies above water table. In some areas yields small quantities of water to wells
			Undifferentiated upland deposits (Qre)		25	Boulders, gravel, sand, and clay	Generally yields small quantities of water to wells, springs, and seeps. Yields moderate quantities from well-sorted gravel in a few places
			Terrace gravel (Qtg)		10	Cobble and pebble gravel; contains sand and silt	Generally lies above water table. In some areas yields small quantities of water to wells and springs where saturated
Mesozoic	Cretaceous	Upper Cretaceous	Arapahoe formation (Ka)		200	Arkosic sand, gravel, and conglomerate interbedded with clay and shale; contains large ironstone concretions	Yields moderate quantities of water to wells. Water of fair to good quality for most uses but dissolved solids concentration varies considerably
			Laramie formation (Kl)		400	Blue-gray silty shale; contains several silty sandstone beds and lignite seams. Coal beds scattered throughout, but thickest and most persistent coal beds occur near bottom	Yields little or no water to wells in report area. Water is of poor quality. Contains hydrogen sulfide, iron, and methane
			Fox Hills sandstone ^{1/} (Kfh)	"B"	80	"Salt and pepper" sandstone uniformly medium-grained, fairly hard, massive	Yields moderate quantities of water to wells. Water of good quality except in areas of local geologic structure, where it may have undesirable amounts of iron, or hardness
				"A"	130	Similar to "B" sandstone above, but has slightly yellow tint. Frequently thin sands interbedded with siltstone and shale, but in places massive	Do
				Milliken	100	Extensive fossiliferous quartz sandstone containing biotite and muscovite. Characteristic yellow color caused by iron oxide	Do
				Transition zone	900	Interbedded sand and shale; becomes more shaly near bottom	Not developed as an aquifer in the report area. May yield small quantities of water of poor quality to wells
			Pierre shale (Kp)		7,500±	Dark-gray to black marine shale, sandy shale, and silty sandstone containing thin limestone lenses. Hygiene sandstone member: Dark greenish-gray and gritty sandstone; about 500 feet thick; lies about 1,500 feet above the base of the Pierre	Yields very small quantities of water to wells from weathered zones. Water is usually of poor quality. Hygiene sandstone member yields small quantities of water to wells
			Niobrara formation (Kn)		300±	Black to gray calcareous marine shale, gray to greenish-white limestone and white chalky marl	Yields very small quantities of poor quality water to wells
			Benton shale (Kb)		500	Blue-gray and dark-brown fissile marine shale, persistent bentonite seams, chalky limestone, and thin sandstone near top	Do
		Lower Cretaceous	Dakota sandstone (Kd)		300	Fine- to medium-grained, friable to firm gray-white sandstone; thin-bedded to massive; contains some shale in the upper part. Dark-gray, silty, carbonaceous shale in the middle part. Coarse-grained gray sandstone; locally conglomeratic in the lower part	Yields small to moderate quantities of water to wells near outcrop
	Jurassic	Upper Jurassic	Morrison formation (Jm)		300	Varicolored claystone, limestone, mudstone, sandstone, and siltstone	No wells tap unit in the report area. May yield small quantities of water to wells from the sandstone beds
			Ralston Creek formation (Jrc)		40	Calcareous sandstone and siltstone in various shades of red and gray	No wells tap unit in the report area
			Entrada sandstone (Je)		30	Fine- to medium-grained well-sorted white sandstone	No wells tap unit in the report area. May yield small quantities of water to wells near outcrop
Paleozoic	Triassic(?)		Lykins formation (A Pl)		550	Interbedded soft sandstone and sandy shales containing some limestone	No wells tap unit in the report area
			Lyons sandstone (Pl)		250	Firmly cemented crossbedded quartz sandstone	Yields moderate quantities of water to wells and springs near outcrop
			Fountain formation (FPf)		1,000±	Crossbedded very arkosic conglomeratic sandstone; contains numerous mudstone and siltstone layers	Yields small quantities of water to wells and springs near outcrop
	Pennsylvanian						
	Precambrian		Crystalline rocks (pc)			Igneous rocks and metamorphosed sediments	Yields small quantities of water to wells and springs from fractured and weathered zones

^{1/} These units are designated as the Milliken sandstone member and the A sandstone and B sandstone of the Laramie formation. For the purpose of the report the sandstone units of the Laramie formation are included as aquifers with the Fox Hills sandstone.

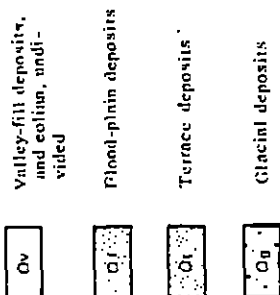
DESCRIPTION OF AQUIFERS NEAR DOWE FLATS

EXPLANATION

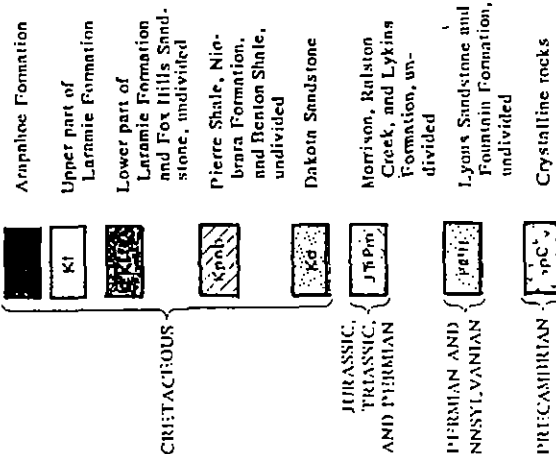
Physical description	Approximate thickness (feet)	Aquifer unit	Well yields ¹
Valley fill--Coarsely bedded, poorly sorted material deposited on slopes and valley bottoms primarily by sheet wash	0-35	Valley fill	Small to inadequate
Eolian--Wind-deposited fine sand and silt	0-25	Flood plain	Small
Stream-deposited boulders, gravel, and sand, with some clay and silt. These deposits are within the flood plain of streams	0-30	Flood plain	Medium to large in plains, medium to small in mountains
Stream-deposited boulders, gravel, and sand, with some clay and silt. These deposits are above the flood plain of streams	0-10	Terrace	Medium
Material of all sizes from boulders to silt. Deposited by glaciers and outwash	0-50	Glacial	Medium to small
Physical description	Approximate thickness (feet)	Aquifer unit	Well yields ¹
Claystone and siltstone with lenses of sandstone	400	Arapahoe	No data
Claystone with some layers of sandstone and coal	600	Upper Laramie	Medium to inadequate
Sandstone with some thin beds of shale	250	Laramie-Fox Hills	Medium
Pierre--Shale with some sandstone beds in the middle and upper part, Hygiene Sandstone Member most prominent	8,000	Pierre-Niobrara-Benton	Inadequate
Niobrara and Benton--Slate with layers of limestone	600		
Sandstone with layers of shale	300	Dakota	Medium to small
Siltstone with beds of sandstone and limestone. Deeply buried except near outcrop	900	Morrison-Ralston-Creek-Lykins	Medium to small
Lyons--Sandstone. Deeply buried except near outcrop	250	Lyons-fountain	Small
Fountain--Conglomeratic sandstone with lenses and layers of siltstone. Deeply buried except near outcrop	1,000		
Fractured igneous and metamorphic rocks	----	Crystalline rock	Small to inadequate

¹For purposes of this report, reported well yields have been classified as follows: Inadequate yields less than 1 gallon per minute; small yields, less than 15 gallons per minute; medium yields, 15 to 100

UNCONSOLIDATED ROCK UNIT

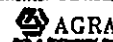


ROCK UNIT



SOURCE: HALL et al (1980)

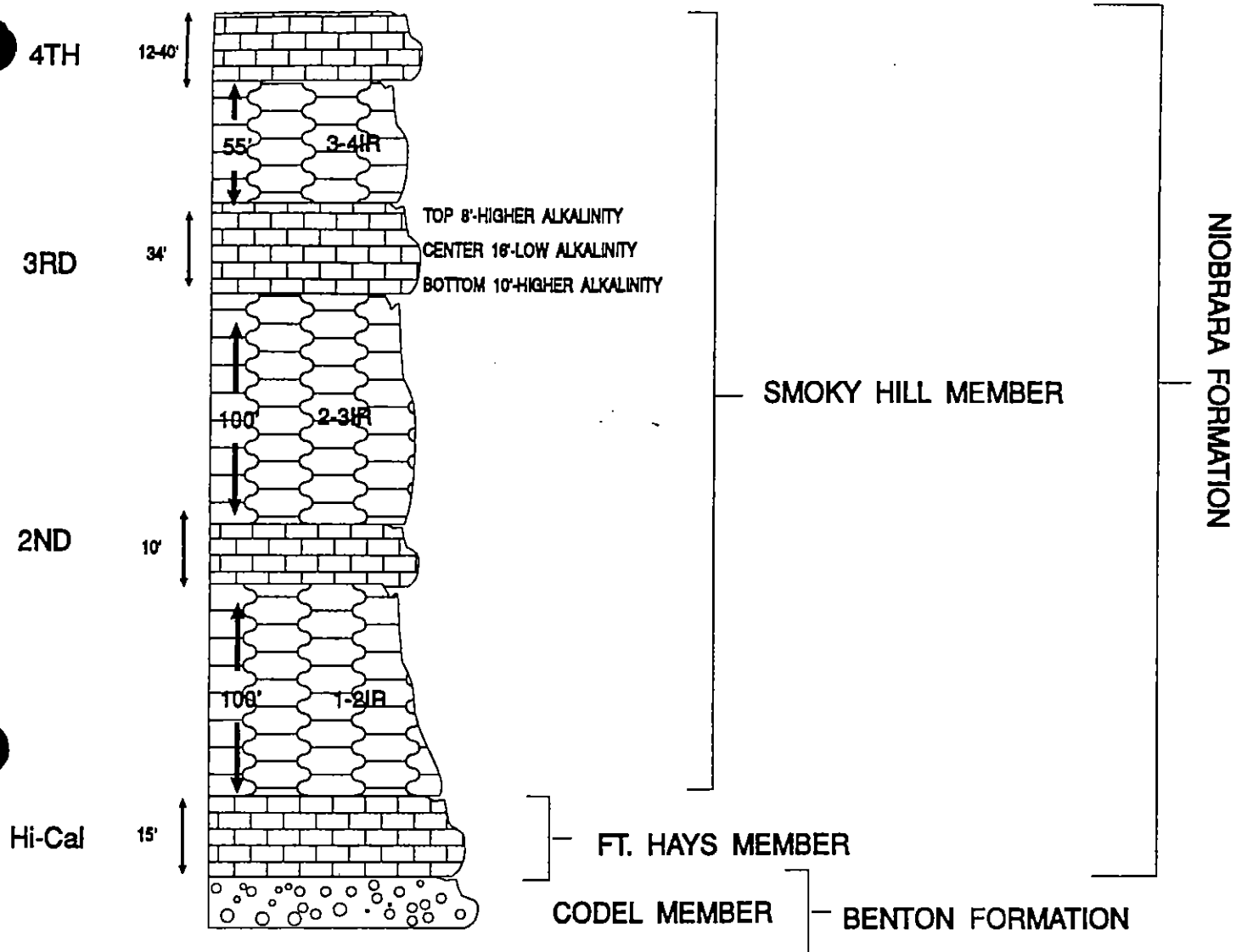
SHB AGRA, INC.
Engineering & Environmental Services



DESCRIPTION OF AQUIFERS

DOWE FLATS PROJECT

SCALE: DATE: 1-21-83 PROJECT NO: E92-7075 FIGURE NO: G-5



SOURCE: WESTERN TECHNOLOGIES INC.



SHB AGRA, INC.
Engineering & Environmental Services



**DETAILED GEOLOGIC SECTION
SHOWING THE NIOBRARA FORMATION
DOWE FLATS PROJECT AREA**

BY: mhb

DATE: 1-21-93

PROJECT No. E92-7075

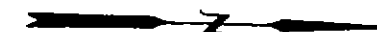
FIGURE: G-6

DFMW-2
5241.12

DFMW-1
5466.74

DFMW-3
5238.99

DFMW-4
5222.15



SCALE: 1"=800'

SOURCE: WESTERN TECHNOLOGIES INC.

SHB AGRA, INC.
Engineering & Environmental Services
AGRA

POTENTIOMETRIC
SURFACE MAP

DOWE FLATS PROJECT

SCALE: 1"=800' DATE: 12-2-80 PROJECT NO.: FIGURE NO.:

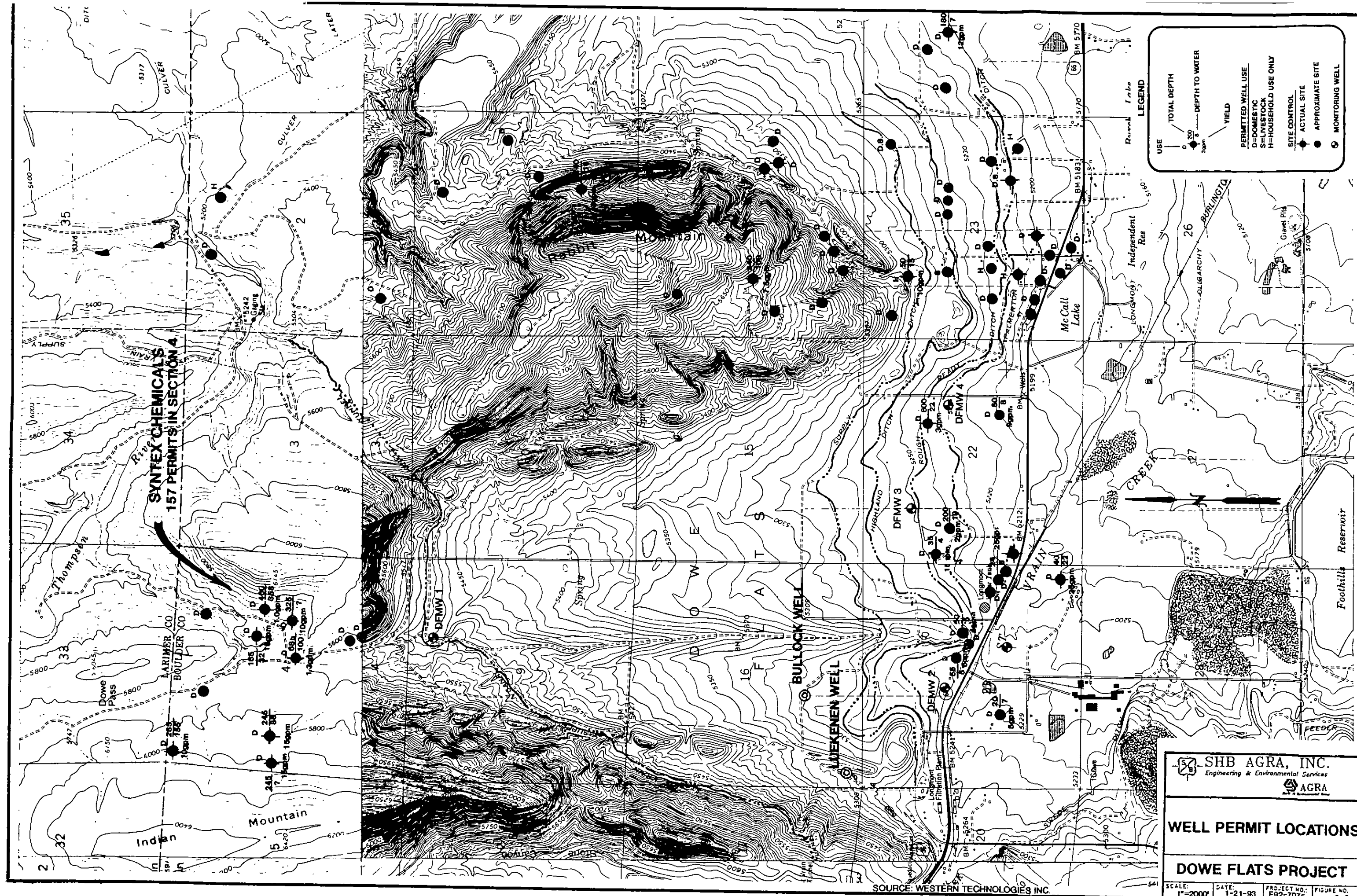


EXHIBIT H - WILDLIFE INFORMATION

Please
provide
the DOW
Comments.

- (a) Description of the significant wildlife resources on and in the vicinity of the application area.

Habitat

Study area characteristics, physiography, habitat and vegetation types present in the Dowe Flats area have been described by ESCO Associates (1987) and Western Ecosystems Inc. (1987).

The marginal croplands and intensively utilized pastures on the permit area support below average wildlife richness and abundance values. However, some of these low quality habitats support large prairie dog towns which are ecologically important because of the prey base they represent, particularly for a large number of wintering raptors.

Relatively undisturbed habitats surrounding the project area including Indian Mountain to the west, Rabbit Mountain to the east, and St. Vrain Creek to the south support rich wildlife communities. Daily and seasonal wildlife movements between these areas cross the project area and add to the importance of the site.

Wildlife Analyses

* stipulations
to approval

Discussions with Boulder County (BC), the Colorado Division of Wildlife (CDOW), U.S. Fish and Wildlife Service (USFWS), and the Colorado Mined Land Reclamation Division (CMLRD), indicate that as a result of Dowe Flats wildlife analyses, studies, and existing data, a wildlife data base exists that is adequate to meet local and state permitting requirements. To complete this data base, a black-footed ferret study (recommended by the CDOW) and a burrowing owl study (recommended by Boulder County) will be conducted prior to any mining. The ferret survey will be conducted between December 31, 1993 and March 31, 1994 following USFWS (1989) - protocol. Three replicated burrowing owl surveys will be conducted between May and August, 1993.

Wildlife Species

Results of Burrowing
owl study complete
? Division requests a
copy

Two primary wildlife issues are associated with the project area: prairie dogs/raptor prey base and big game.

1. The prairie dog/raptor issue is clearly the most ecologically important wildlife

issue associated with the project. Holistic Resource Management (1987) located 3,142 active prairie dog burrows on approximately 869 acres in 1987 and prairie dogs have expanded their distribution in recent years. This colony is important to relatively large numbers of bald eagles (a state and federal endangered species) and ferruginous hawks (a federal candidate species) that winter in and around Dowe Flats. Other terrestrial and avian predators also exploit this prey base. Associated with this issue are the eagle and hawk roost/hunting perch sites in cottonwoods flanking irrigation ditches.

2. Big game, including mule deer, white-tailed deer, and elk, are seasonally present in and around the project area. Mule deer are of greatest concern, since they utilize Dowe Flats to a greater extent than the other big game species. Mule deer use Dowe Flats as an east-west movement corridor and, to a limited extent, as winter range.

Other sensitive wildlife-related issues associated with the project area include the riparian corridor and nongame values along the St. Vrain River, potential destruction of raptor nest trees, and general loss of wildlife habitat.

(b) Seasonal wildlife use of the area

The greatest wildlife value of Dowe Flats is that of winter habitat (November through March) for a relatively large number of raptors, including bald eagles, golden eagles, ferruginous hawks, red-tailed hawks, rough-legged hawks, northern harriers, prairie falcons, and American kestrels. The first three species are particularly abundant and are attracted to the area by the large number of prairie dogs.

Mule deer use Dowe Flats for winter range to a limited extent. Elk winter range overlaps the western and northern portions of the permit area, but these areas of intensively utilized pasture and rotated agricultural fields are generally fallow in winter and are not used by elk during most years. White-tailed deer in the vicinity of the permit area are generally associated with the riparian habitats flanking the irrigation ditches to the south of the permit area. Accordingly, the basic big game mitigation assumption is that if the seasonal needs of mule deer can be met on and through the permit area, those of elk and white-tails will also be protected.

(c) Presence and estimated population of threatened or endangered species from either federal or state lists

The Dowe Flats area supports a relatively large number of bald eagles (a state and federal endangered species) and ferruginous hawks (a federal candidate species) during the winter, and impacts to their prey base (prairie dogs) will be mitigated. There is no documented evidence to suggest that any other terrestrial state or federal threatened, endangered, or candidate species, including the whooping crane, sandhill crane, peregrine falcon, interior least tern, piping plover, white-faced ibis, western snowy plover, mountain plover, long-billed curlew, Preble's meadow jumping mouse, black-footed ferret, or swift fox are present within the permit area. Surveys for black-footed ferrets will be conducted in Dowe Flats during winter 1993/94 following USFWS (1989) protocol.

(d) Description of impacts caused by the proposed operation

Description of Impacts and Reclamation Effects

This section describes impacts to wildlife during and after mining and reclamation operations. Wildlife issues were originally described by Western Ecosystems Inc. (1987). Description of impacts to wildlife takes into consideration the effects of implementation of avoidance, minimization, mitigation and enhancement measures outlined in the reclamation plan. Discussions are detailed for those wildlife species and groups that are of special concern, and are less specific for other wildlife groups.

What about during mining operations - before reclamation?

Raptors

Although a wide variety of raptors seasonally hunt Dowe Flats, specific concerns focus on bald eagles and ferruginous hawks because of the permit area's importance to these endangered and candidate species. Mining impacts to raptors will be entirely indirect, as the result of reduced prey base availability. This reduction will be most significant during winter (November through March) when relatively large numbers of these birds focus on the permit area's large prairie dog towns. Therefore, the focus of the raptor issue is the availability of prairie dogs as prey base for raptors throughout the mining process.

Prairie dog towns (Exhibit H Map) in the project area were initially documented by Holistic

6 years ago current enough to be accurate?
Resource Management (1987). All acreages used in this section are approximate due to the dynamic nature of prairie dog distribution and a conservative estimate of the acreage of reclaimed areas that will be suitable for prairie dog habitat.

Prairie dog distribution in the valley is as widespread as it has ever been in recent years, due in part to reduced prairie dog control efforts by the present land managers. Prairie dogs have been controlled for decades in this valley by poisoning, plowing, shooting, and by other methods. As a result of the recent reduction in prairie dog control efforts, and due to the fact that Dowe Flats has avoided recent plague epizootics which periodically decimate local populations, prairie dogs are presently distributed throughout most of the valley.

Prairie dog distribution has recently expanded from four towns in the valley, and now totals at least 1,547 acres across the bottom of Dowe Flats. The largest town, running the length of the valley, totals at least 800 acres and is the only town that will be directly or indirectly affected by mining, reclamation, and road realignment. Over the 25 year life of mining activity, much of this town will be affected by mining, although no more than 12% of it will be affected at any given time.

How much? acreage?

As mining proceeds up the valley, reclamation would follow. Reclaimed landforms, specifically designed to accommodate prairie dogs and raptors, would be rapidly colonized as early successional vegetative communities provide ideal prairie dog habitat. A minimum of 30% of the affected acres will be reclaimed as habitat suitable for prairie dogs, lagomorph and deer habitats. Additional reclamation features (e.g. rimrocks) and habitats would be oriented toward enhancing lagomorph and deer habitats, additional prey species for the large raptors.

As mining begins and progresses, individuals in portions of the large prairie dog town would be killed or would colonize adjacent unoccupied habitats. Raptors hunting these towns for prairie dogs and other resident prey species may decrease in number due to direct and indirect influences of mining activity, including reduced prey availability, avoidance of prey in habitat adjacent to mining and human activity, and the disturbance and temporary abandonment of roost sites, hunting perches, and hunting areas adjacent to haul roads and the relocated county road.

Total direct impact in acres?

Post-reclamation prairie dog acreage in the valley would total 75% of that now present. Additional suitable, but presently unoccupied adjacent habitats within the permit area, would also be available for colonization during or after the mining process, but are not included in these

✓
they say
raptor
habitat
enhancement
substitutes
for prairie
dog towns loss

totals. Due to reclamation plan design, prairie dog colonies will also be spatially isolated from each other to a greater degree than at present, thereby reducing the probability that a plague epizootic, which routinely decimates prairie dog towns along the Front Range, would wipe out the entire raptor prey base in Dowe Flats.

In addition to the perpetuation, specific reclamation, and isolation of large prairie dog towns, implementation of the raptor mitigation and enhancement measures discussed below are expected to more than compensate for any values lost during the mining process. These measures are summarized in the Mitigation section of this document.

Raptor Habitat Mitigation and Enhancement Measures

- Avoidance of the bald eagle roost/hunting perches in the mature-decadent cottonwoods flanking the irrigation ditches east of the present county road (Exhibit H Map, 6200 to 8400 north X 7700 to 10,700 east)

isn't this
an imbalance
for this
area -> this
for many
raptors for
prey base?

Establishment of a large cottonwood stand and numerous smaller stands, that could eventually serve as a bald eagle roost. Although only a few cottonwoods would be removed from the haul road corridor crossing the ditches, it is expected that vehicular disturbance could displace eagles from a zone up to a maximum of 100 yards of each side of these roads (a 200 yard corridor). This distance is based on observations of perched, roosting, and hunting bald eagles adjacent to similar active mining operations at Southwestern's Lyons mine, approximately two miles south of Dowe Flats. It is expected that portions of the present roost will remain viable, but its capacity will be reduced. To enhance future roost availability in Dowe Flats, a replacement cottonwood stand would be established within the reclamation mosaic in an area visually screened (minimum 0.25 mile buffer) from potential disturbances outside the reclaimed landscaping (Exhibit F - Reclamation Plan Map).

- Installation of raptor hunting perches throughout the affected area. Initially, telephone poles will be established atop reclaimed landforms to facilitate aerial predation. Poles would be established along the perimeter and within the interior of the reclaimed areas overlooking prairie dog towns, reclaimed rimrock areas, and most habitats that are expected to be colonized by prairie dogs. Eventually,

cottonwood saplings and ponderosa pines planted in shallow drainage basins atop and between landforms will naturally replace the function of the artificial poles. Bald eagles currently roost in and hunt from cottonwoods in an existing mine pit, south of the Southwestern plant.

- Creation of landforms/water features with diverse habitats supporting other (non prairie dog) prey species (e.g., small rodents, lagomorphs, nongame birds, and waterfowl). The additional raptor hunting opportunities provided by the wildlife prey base in these habitats could compensate for reduced prairie dog availability.
- Creation of rock outcrops (rimrock) for lagomorphs, potential prey species for bald and golden eagles, ferruginous hawks, and other raptors.
- Retention of 2500 linear feet of 100 to 120 foot tall, unreduced highwalls in appropriate areas to provide suitable sites for cliff-nesting raptors. No such cliff nesting habitat presently exist in Dowe Flats. Such cliffs, surrounded by proposed, complementary landforms/vegetation types, and visually screened from most the Dowe Flats valley bottom, could realistically be occupied by great horned owls, American kestrels, prairie falcons, golden eagles, ferruginous hawks, and red-tailed hawk, barn owls and common ravens. Depending on final highway characteristics, "potholes" could be created (1 per 100 linear feet of cliff) to create additional nesting ledges. Other raptors, such as long-eared owls and sharp-shinned and Cooper's hawks, could also nest in adjacent habitats. These cliffs would also be used as nest sites for potential prey species (e.g., rock doves and cliff swallows) and other wildlife ecologically important to the community.

*Are the highwalls
at the Lyons
 Quarry inhabited
by these
species?*

No burrowing owls have been observed in Dowe Flats. A systematic survey for burrowing owls will be conducted between May and August 1993 to confirm their absence. If burrowing owls are observed, appropriate mitigation measures, that are acceptable to Boulder County and the proponent, will be cooperatively developed and implemented.

Big Game

Although mule deer, white-tailed deer, and elk are seasonally present in the vicinity of the

permit area, big game concerns focus primarily on mule deer because of the relative importance of the permit area to this species as a broad, east-west movement corridor that presently extends along the length of the valley. Important winter ranges on Rabbit Mountain are accessed by deer moving east across Dowe Flats in the fall. This pattern is reversed in spring as some deer migrate west to higher elevation summer ranges.

Winter range values of Dowe Flats are marginal because native vegetation is heavily utilized by cattle and prairie dogs, leaving little standing biomass during winter, and because agricultural crops are generally fallow for the winter. Mule deer, which use portions of Dowe Flats during winter, take pressure off the heavily utilized Rabbit Mountain winter range. The intensity of utilization on Rabbit Mountain has been exacerbated in recent years by an increasing deer population and the CDOW's difficulty in managing this herd because the area and surrounding lands are closed to all hunting or to public hunting.

The proposed mining will have little adverse effect on the local deer herd. Although deer would be locally restricted from active pits and from areas awaiting reclamation, local and seasonal movements across Dowe Flats would continue. Broad corridors north and south of active mining and within the existing corridor would be more than adequate to accommodate continued movements (Exhibit F - Reclamation Plan Map).

The only obstacles that will restrict east-west deer movements from portions of their present corridor would be unreduced highwalls and ponds/lakes. However, neither of these features would adversely affect deer movements, because configuration of reclaimed landforms in these areas have been specifically designed to facilitate east-west movements (Exhibit F - Reclamation Plan Map). Valleys, buffered from the surrounding area by ridges, and vegetated with valuable big game foraging and cover species, should allow for leisurely, undisturbed residency as well as movements. The present, existing corridors also extend north and south of the permit area, so it would not be necessary to accommodate all future movements within the permit area. If fencing is constructed to restrict deer from reclaimed areas until after the vegetation could withstand big game grazing, east-west corridors would be provided between large fenced reclaimed areas to allow for continued movements.

Mining activity is not expected to increase traffic volumes on County Road 47, and no increase is expected in big game highway mortalities along the relocated County Road 47 or Highway 66. Haul truck activity between the mine and plant would be at 35 mph or less, speeds at which

How monitored?

✓
Division would require fencing of reclaimed areas until Reg. is established

big game-vehicle collisions are infrequent. The movement corridor south of the proposed mine is sufficiently wide through unrestricted habitats to allow for continued movements. Disturbances to deer in this corridor will occur primarily from nonmine-related activities and are not expected to force deer south onto Highway 66.

After mining and reclamation, Dowe Flats would support much more valuable seasonal big game habitats than what are now present. Diverse landforms will support a variety of native vegetation types (e.g., grasslands, shrublands, herbaceous and shrubby wetlands, thickets and groves of trees) seasonally important to big game and other wildlife groups (see reclamation plan). Buffering landforms, cover, forage, and additional water sources will not only facilitate and enhance movements and winter range values, but also provide suitable fawning, transitional, and summer habitats. It is likely that the post-mining community would support a resident deer herd.

It is recommended that some form of hunting be permitted within the permit area to facilitate deer and habitat management.

Other Wildlife Groups

Other wildlife groups, including small mammals, nongame birds, waterfowl and shorebirds, terrestrial predators, raptors, and fish, will experience minor impacts during mining, but will significantly benefit from the highly productive habitats that are created following mining. Although mitigation and enhancement measures will focus on the needs of raptors and big game, post-mining conditions and opportunities will be exploited by other wildlife groups, greatly improving present levels of wildlife diversity and abundance.

In addition to previously mentioned mitigation measures, landforms (e.g., convoluted ridges, incised valleys, creeks, ponds, cliffs, islands, and gently rolling topography) will be created that support a diversity of vegetation types and a correspondingly rich wildlife community. Components within these landforms will include grasslands, shrublands, herbaceous and shrubby wetlands, thickets and groves of trees, nesting cliffs/drop-offs for fisheries, spawning beds, artificial reefs, meandering creeks, waterfowl nesting/roosting islands, and diving and dabbling duck foraging areas. The goal of reclamation within the former mine pits is to create an ecological community and setting similar to that of White Rocks along Boulder Creek, considered to support one of the most diverse and unique wildlife habitats in Boulder County.

Representatives of some wildlife groups now inhabiting the area will be displaced from localized portions of Dowe Flats for the life of the mining effort. Moreover, some species (e.g., badger, ring-necked pheasant, and vesper sparrow) may have a slightly smaller acreage of suitable habitat following mining and reclamation than what now exists. However, the greater diversity and abundance of wildlife supported by the post-mining habitats should outweigh the reduced abundance of species with affinities toward more common habitats of pasture and marginal cropland.

Rule 2.1.2 (8) (b) (c) ~~added~~

— show how water from runoff from the disturbed areas, piled material and operating surfaces will be managed to protect against pollution of either surface or ground water (Stormwater Discharge Permit)

— provide estimate of the project water requirements including flow rates and annual volumes for development, mining, and reclamation phases of the project

Perman
included
in
hydrology
permit

EXHIBIT I - SOILS INFORMATION

Soils of the Dowe Flats project area were mapped by the U.S. Soil Conservation Service in 1975 as part of the Boulder County Soil Survey. The three main units mapped in areas that will be disturbed by mining include LaPorte very fine sandy loam, Manvel loam, and Gaynor silty clay loam (Exhibit I Map).

The LaPorte soils are shallow and well-drained, developing over limestone and limy shale. They are possessed of moderate permeability, but because of their shallowness have low water-retention capacity. The upper horizon on the Dowe Flats site is a dark brown sandy loam from 4 to 7 inches thick, somewhat thinner than described in SCS (1975) where the upper horizon was described as 8 inches thick. Underlying is a layer of pale brown calcareous loam about 4 inches thick. Depth to parent material varies from 11 to 13 inches. These soils are present much as mapped, except where they extend into areas mapped as Gaynor (see below); thickness of the A horizon has varied from 4 to 10 inches.

Manvel soils are deep and well-drained, developing on calcareous, loamy alluvium. Permeability in this soil is moderate and in conjunction with the great soil depth, water holding capacity is low. The upper layer is a brown loam about 6 inches thick. The underlying layer is a pale brown loam about 14 inches thick. Below this, a dark grayish brown to dark brown silty clay loam extends to about 60 inches. This soil covers the broad central areas of Dowe Flats, away from limestone outcrops and away from the slopes of Rabbit Mountain.

The Gaynor soils are moderately deep soils derived from loamy alluvial and wind-blown materials. They are mapped in areas expected to have been the site of wind-deposits, sites rather difficult to predict exactly. Much of the area mapped as Gaynor is either LaPorte or Manvel series. The surface layer is a grayish brown silty clay loam about 6 inches thick. Below this is a brown silty clay loam about 4 inches thick. The underlying layer of silty clay loam extends another approximately 20 inches deep.

Soil samples have been submitted to the Colorado State University Soils Laboratory for analysis to verify reclamation suitability. Slight modifications to the boundaries of soil units have been made but are not considered critical because all material is anticipated to be rated as suitable for revegetation, and all units are likely to be salvageable to similar depths.

Other Soils of the Permit Area

Soil units within the permit area that are not to be disturbed or which will experience only very minor disturbance include Ascalon sandy loam, Ascalon-Otero complex, Baller stony sandy loam, Calkins sandy loam, Loveland soils, Nunn sandy clay loam, Nunn clay loam, Valmont clay loam, and Valmont cobbly clay loam.

Ascalon soils are fine loamy, mixed, mesic Aridic Argiustolls, deep and well-drained, formed on terraces and uplands in loamy mixed alluvium and wind-laid materials. A sandy loam surface horizon of about 8 inches is underlain by a textural B-horizon of about 11 inches; below this, a calcareous subsoil extends to a depth of 60 inches or more.

Baller soils are loamy-skeletal, mixed, mesic Lithic Haplustolls, shallow and well-drained, formed on upland ridges and slopes in loamy residuum weathered from sandstone. Typically, the surface layer is a grayish brown stony sandy loam about 10 inches thick. Below this, 5 inches of light brownish gray very stony sandy loam overlies the sandstone parent material.

Calkins soils are coarse loamy, mixed, mesic Cumulic Haplaquolls, deep and somewhat poorly drained, formed from loamy alluvium on low terraces and bottomlands. Typically, the surface layer is a grayish brown sandy loam extending to a depth of 40 inches. Below, a light-brownish gray coarse sandy loam extends to 60 inches or more.

Loveland soils are fine loamy over sandy or sandy-skeletal, mixed (calcareous), mesic Typic Haplaquolls, deep and somewhat poorly drained, formed on loamy alluvium overlying sandy and gravelly material on terraces and bottomlands. Typically, the surface layer is a calcareous dark grayish brown light clay loam about 20 inches thick, mottled in the lower parts. Below this a layer of gypsic and strongly calcareous mottled light clay loam about 10 inches thick overlies mottled light brownish-gray gravelly sand extending to a depth of 60 inches or more.

Nunn soils are fine montmorillonitic, mesic Aridic Argiustolls, deep and well-drained, formed on terraces and valley side slopes in loamy alluvium. Typically the surface layer is a grayish brown clay loam about 10 inches thick. The subsoil is a brown and very pale brown clay, grading to clay loam, about 10 inches thick. Below this, the substratum is a calcareous very pale brown clay extending to 60 inches or more.

Otero soils are coarse loamy, mixed (calcareous), mesic Ustic Torriorthents, deep and well-drained, formed on terraces, rolling uplands, and valley slopes in loamy alluvium and wind-laid materials. Typically, the surface layer is a strongly calcareous brown sandy loam about 6 inches thick. The subsoil is a pale brown sandy loam about 6 inches thick. The substratum material is a strongly calcareous light yellowish brown sandy loam, extending to a depth of 60 inches or more, with lime concretions in the upper parts.

Valmont soils are clayey over loamy-skeletal, montmorillonitic, mesic Aridic Argiustolls, deep and well-drained, formed on old, high terraces in cobbly and gravelly loamy alluvium. The surface layer is a grayish brown light clay loam, with varying amounts of gravel and cobbles, about 4 inches thick. An upper subsoil layer of brown clay loam about 3 inches thick overlies a middle subsoil layer of brown light clay about 13 inches thick. The lower subsoil is a calcareous light brown gravelly clay loam about 4 inches thick. Below this, a layer of pinkish-white and light brown very gravelly loam extends to a depth of 60 inches or more.

Impacts on Soils and Vegetation

The vegetation of the Dowe Flats area that will be disturbed by the proposed mining operation is largely agricultural, either currently in cultivation, or having previously been plowed and planted to European or Asian forage species. What remains from the native vegetational cover is present on the approximately 75 acre crest and upper slopes of "Limestone Ridge". This area has a substantial array of native plant species, though none considered rare, threatened or endangered.

During the course of mining, the vegetation of the site will be removed, so that the existing value it has as an agricultural resource and as wildlife habitat will be absent. Upon regrading and the onset of reclamation, vegetation will be restored to the site that will comprise far more diverse biological communities, occurring in more diverse topographic settings than exists presently, limited as they are by the present homogeneous, nearly flat topography and a history of biological simplification for the purpose of improved agricultural utility.

Soils of the site are not suitable for cultivation due to erosion hazard and slow permeability; nonetheless they have been cultivated historically, and have suffered some degradation of surface horizon thickness as a consequence. The highest use of these soils is in support of a cover of perennial vegetation, whether for support of livestock grazing, or wildlife habitat, or both.

Soil amend
like Nitrogen
(see next page)?

Where?

How will
they be
stabilized?

CMLRB - Exhibit I
Soil Information
Page 4

During mining, these soils will be sequentially removed, sometimes stored temporarily, and replaced, then revegetated to a perennial vegetational cover. The existing configuration of soil horizons in the form of soil profiles will be lost during salvage, but upon reclamation, this soil material will be overlain by a protective mostly native vegetational cover that will allow soil development to again proceed to a degree that it has been unable to during its most recent history of agricultural use and abuse.

Mitigation of Impacts

Upon the initiation of reclamation, the existing degraded vegetational and soil resources of Dowe Flats will be replaced and incorporated, respectively, by/into a landscape of far greater topographic and vegetational diversity than exists now. Vegetation types will include upland grasslands and shrublands, wet meadows and marshes, as well as open water, and moist shrubland and forests in protected coves. All will be integrated into a complex mosaic that will provide much more diverse plant and wildlife habitat, as well as pleasing aesthetic qualities.

Soil Sample*	Lab No.	pH (paste)	E.C.** (paste)	Lime (est.)	Org. Mtr. (%)	NO3-N (ppm)	P (ppm)	K (ppm)	Zn (ppm)	Fe (ppm)	Mn (ppm)	Cu (ppm)	Texture (CSU)	Texture (ESCO)
Manvel A 0-6 in.	R4705	7.7	0.8	High	2.2	5	9.9	495	2.0	7.6	7.1	2.9	scl	vfsi
Manvel B 0-4 in.	R4706	7.6	0.4	High	2.4	5	0.3	166	0.4	6.0	1.3	3.0	cl	l
Manvel B 4-8 in.	R4707	7.7	0.4	High	2	4	<0.1	153	0.4	11.3	1.6	4.0	cl	hl
Manvel B 8-16 in.	R4708	7.7	0.4	High	1.8	5	0.5	143	0.5	11.5	1.5	3.4	cl	fsi
Manvel B 16-20 in.	R4709	7.8	0.4	High	2.2	4	0.8	42	0.5	7.3	0.7	2.6	scl	sil
Gaynor A 0-4 in.	R4710	7.5	1.2	High	3.7	35	2.5	758	0.9	9.9	4.7	2.5	scl	l
Gaynor A 4-8 in.	R4711	7.5	0.7	High	2.4	6	3.8	465	0.9	8.3	4.9	2.7	scl	l
Gaynor A 8-12 in.	R4712	7.6	0.6	High	1.7	7	0.3	212	0.4	7.9	1.5	2.2	cl	l
Gaynor B 0-6 in.	R4713	7.7	0.6	High	2.3	8	13.5	458	0.9	6.7	2.2	2.5	scl	fsi
Gaynor C 0-6 in.	R4714	7.7	0.7	High	2.3	17	12.7	363	0.9	8.4	4.1	2.2	scl	fsi
LaPorte A 0-6 in.	R4715	7.6	0.7	High	3.2	8	0.5	555	0.4	9.3	1.4	2.6	scl	sl
LaPorte B 0-6 in.	R4716	7.7	0.6	High	3	8	3.1	363	1.2	10.3	2.8	4.0	scl	sl
LaPorte C 0-6 in.	R4717	7.6	0.5	High	2.4	7	0.8	263	0.6	10.9	2.0	3.9	scl	sl
LaPorte C 6-12 in.	R4718	7.7	0.4	High	1.6	3	<0.1	123	0.3	12.2	1.1	2.6	c	fsi
LaPorte C 12-18 in.	R4719	7.7	0.4	High	1.5	3	<0.1	116	0.3	12.1	1.0	2.6	c	fsi
LaPorte C 18-24 in.	R4720	7.8	0.3	High	1.1	3	<0.1	118	0.3	12.0	0.8	2.4	c	fsi
LaPorte D 0-4 in.	R4721	7.6	0.6	High	3	6	2.5	277	0.8	7.7	3.0	2.2	scl	sl
LaPorte D 4-8 in.	R4722	7.6	0.5	High	2.5	6	0.3	130	0.3	5.9	1.9	2.2	cl	fsi
LaPorte D 8-12 in.	R4723	7.6	0.4	High	1.9	3	0.5	104	0.3	6.6	1.4	2.0	cl	fsi
Overburden "clay"	R4724	7.9	1.7	High	0.6	2	1.7	53	0.6	7.9	1.3	1.4	c	cl
Overburden "rock"	R4725	7.7	4.8	High	2.5	4	<0.1	105	15.7	24.2	1.1	2.3	scl	l

* Soil sample location shown on Map I-1.

** Electrical conductivity (mmhos/cm)

Soil analyses conducted by Colorado State University Soil Testing Laboratory, Fort Collins.

Textures as estimated by Colo. St. Univ. Lab. technicians (CSU) and during field sampling (ESCO).

Texture abbreviations: scl - sandy clay loam; cl - clay loam; c - clay; vfsi - very fine sandy loam; fsi - fine sandy loam; sil - silt loam; sl - sandy loam; hl - heavy loam.

Parameter	Suitable			
	Good	Fair	Poor	Unsuitable
pH	6.0-8.4	5.5-6.0 8.4-8.8	5.0-5.5 8.8-9.0	<5.0 >9.0
E.C. (mmhos/cm)	0-4	4-8	8-16*	>16
Texture	sl, l, sil, scl, vfsl, fsl	cl, sld, sc, ls, lfs	c, slc, s	

* >8 may prove difficult to revegetate.

Source: Wyoming Department of Environmental Quality. 1981. Guideline No. 1,
Topsoil and Overburden.

EXHIBIT J - VEGETATION INFORMATION

The mapped vegetation types of the Dowe Flats permit area are shown on the Exhibit J Map. These were mapped from aerial photos (U.S. Geological Survey NHAP 1983, scale - 1:40,000).

Vegetation of the Area to be Disturbed

Vegetation of the area to be disturbed by the mining is comprised of four types:

- Improved pasture
- Hay pasture
- Cultivated land
- Grassland along "Limestone Ridge"

Improved Pasture areas were probably once cultivated and were then planted to Russian wildrye (Elymus junceus). These areas are heavily used by cattle and extensively occupied by black-tailed prairie dogs. Between these animals, overgrazing is pronounced and, to varying degrees, various other plants of weedy character have invaded these lands. Invaders include snakeweed (Xanthocephalum sarothrae), musk thistle (Carduus nutans), Canada thistle (Cirsium arvense), common bindweed (Convolvulus arvensis), and the native plains prickly poppy (Argemone polyanthemus).

Hay Pasture is irrigated land that has been planted to such domestic forage grasses as smooth brome (Bromopsis inermis), orchard grass (Dactylis glomerata), common timothy (Phleum pratense), and/or tall fescue (Festuca arundinacea). These lands are mowed for hay during the growing season and may later experience grazing after the growing season.

Cultivated Land is mostly managed for the production of small grains, especially winter wheat and oats, but occasionally corn. Most of it is handled in a strip rotation pattern; periodically, alfalfa is rotated in to improve soil fertility and tilth.

The Grassland along "Limestone Ridge" is a remnant of the original vegetation cover of Dowe Flats. Although it has experienced heavy disturbance through grazing and other activities, it retains a diversity of native plants not found in the more extensive created pastures and cultivated land adjacent. Dominant species in this stand are the perennial native grasses blue grama (Bouteloua gracilis), western wheatgrass (Agropyron smithii), needleandthread (Stipa comata), bottlebrush squirreltail (Sitanion longifolium), and Fendler three-awn (Aristida fendleriana), along with the non-native intermediate wheatgrass (Agropyron intermedium), standard crested wheatgrass (Agropyron desertorum), smooth brome, and Russian wildrye. Introduced annual grasses are locally prominent, and include cheatgrass (Bromus tectorum), which is especially abundant, as well as Japanese brome (Bromus japonicus), annual ryegrass (Secale cereale), and wheat (Triticum aestivum). Native annual grasses present in very small amounts are sixweeks fescue (Vulpia octoflora) and false buffalograss (Munroa squarrosa).

Shrubs are scattered sparsely but rather uniformly and include rubber rabbitbrush (Chrysothamnus nauseosus), winterfat (Krascheninnikovia lanata), three-leaf sumac (Rhus trilobata), prickly rose (Rosa acicularis), and snakeweed. The numerous forbs present include textile onion (Allium textile), prickly poppy, fringed sage (Artemisia frigida), white prairieaster (Aster ericoides), Porter aster (Aster porteri), slender milkvetch (Astragalus gracilis), three-leaf milkvetch (Astragalus tridactylus), downy paintbrush (Castilleja sessiliflora), mouse-ear chickweed (Cerastium arvense), James hiddenflower (Cryptantha jamesii), wafer-parsnip (Cymopterus acaulis), white prairie clover (Dalea candida), low fleabane (Erigeron pumilus), slenderbush wildbuckwheat (Eriogonum effusum), robust spurge (Euphorbia robusta), wallflower (Erysimum asperum), evolvulus (Evolvulus nuttalliana), blanketflower (Gaillardia aristata), scarlet gaura (Gaura coccinea), ironplant goldenweed (Haplopappus spinulosus), stiff sunflower (Helianthus rigidus var. subrhomboideus), hairy goldenaster (Heterotheca villosa), thread-leaved hymenopappus (Hymenopappus filifolius), low bladderpod (Lesquerella montana), sandlily (Leucocrinum montanum), spotted gayfeather (Liatris punctata), Lewis flax (Linum lewisii), salt-and-pepper (Lomatium orientale), bractless blazingstar (Mentzelia nuda), yellow blazingstar (Mentzelia speciosa), lanceleaf chimingbells (Mertensia lanceolata), musineon (Musineon divaricatum), false dandelion (Nothocalais cuspidata), yellow stemless evening primrose (Oenothera brachycarpa), cutleaf evening primrose (Oenothera cornopifolia), Colorado loco (Oxytropis lambertii), James nailwort (Paronychia jamesii), sidebells penstemon (Penstemon secundiflorus), plains bahia (Picradeniopsis oppositifolia), narrowleaf scurfpea (Psoralea tenuiflora), Britton skullcap (Scutellaria brittonii), lambstongue groundsel (Senecio integerrimus), broom groundsel (Senecio spartioides), stiff goldenrod (Solidago rigida), cowboy's delight

(Sphaeralcea coccinea), Rio Grande greenthread (Thelesperma megapotamicum), Easter daisy (Townsendia cf. hookeri), and Dakota vervain (Verbena ambrosifolia).

Introduced plants present in the "Limestone Ridge" grassland include field bindweed, black medic (Medicago lupulina), alfalfa (Medicago sativa), and common dandelion (Taraxacum officinale).

Annual and biennial forbs and grasses are abundant in this area. Most are introduced; these include alyssum (Alyssum minus), redroot pigweed (Amaranthus retroflexus), littlepod falseflax (Camelina microcarpa), musk thistle, diffuse knapweed (Centaurea diffusa), purple mustard (Chorispora tenella), pinnate tansymustard (Descurainia pinnata), flixweed tansymustard (Descurainia sophia), filaree (Erodium cicutarium), common sunflower (Helianthus annuus), summercypress (Kochia iranica), stickseed (Lappula redowskii), false salsify (Podospermum laciniatum), Russian-thistle (Salsola iberica), Jim Hill mustard (Sisymbrium altissimum), salsify (Tragopogon dubius), and bracted verbena (Verbena bracteata). Native annual and biennial forbs present include thyme-leaf spurge (Chamaesyce serpyllifolia), gray goosefoot (Chenopodium incanum), yellow-center thistle (Cirsium ochrocentrum), wavy-leaf thistle (Cirsium undulatum), hare's ear (Conringia orientalis), Carolina draba (Draba reptans), Ellisia (Ellisia nyctelea), snow-on-the-mountain (Euphorbia marginata), plains sunflower (Helianthus petiolaris), Virginia pepperweed (Lepidium virginicum), dragonhead (Moldavica parviflora), and black nightshade (Solanum americanum).

Other Vegetation Types of the Permit Area

Ponderosa Pine Woodland

This mapping unit is dominated by ponderosa pine (Pinus ponderosa) which provides from about 30 to 100 percent canopy cover. The understory is variously comprised of native grasses and shrubs. The mapping unit grades into Mountain Mahogany Shrubland, and, thus may have extensive dense mountain mahogany (Cercocarpus montanus) shrub understory.

Skunkbrush, American Plum, Chokecherry Shrubland

In these areas, deep colluvial deposits support thickets of skunkbrush (Rhus trilobata), American plum (Prunus americana), and Chokecherry (Prunus virginiana var. melanocarpa). Occasional squaw currant (Ribes cereum) and even stunted hackberry (Celtis occidentalis) may be found in these thickets. The American plum and chokecherry do not achieve their typical heights, staying mostly below one meter. This is attributable to the fact that they occur on south- and west-facing slopes that are naturally relatively dry and hot.

Riparian Shrubs and Small Trees

Occurring in narrow strips along a few sheltered drainages are developments of mesic shrubs including American plum and chokecherry as well as willows (Salix sp.) and hawthorns (Crataegus sp.) that reach heights of a few meters. A few tree species including boxelder (Acer negundo), cottonwood (Populus deltoides and P. X acuminata), and peachleaf willow (Salix amygdaloides) occur sporadically.

Dense Mountain Mahogany Shrubland

This extensive mapping unit is distinguished by the heavy cover of mountain mahogany. In some parts, there are scattered ponderosa pine. Herbaceous understory consists of sparse cover of species mentioned in the next unit.

Grassland with Scattered Mountain Mahogany

This grassland unit is characterized by the occasional occurrence of mountain mahogany shrubs, mainly on microsites of shallow soil, where grass competition is limited. The grass matrix is dominated by a mixture of cool and warm season species such as western wheatgrass (Agropyron smithii), needle-and-thread (Stipa comata), junegrass (Koeleria macrantha), sideoats grama (Bouteloua curtipendula), blue grama (Bouteloua gracilis), little bluestem (Schizachyrium scoparium), and Indiangrass (Sorghastrum nutans). Locally, the introduced annual grasses Japanese brome and cheatgrass are overwhelmingly dominant.

Grassland

Over extensive portions of the north and west parts of the study area, grassland dominated by the above species occurs. In general, locations further upslope (above the agricultural activities of the valley bottom) are less modified and less likely to be dominated by Japanese brome or cheatgrass. Some areas of grassland are heavily infested by noseburn (Tragia urticifolia).

Wetland

These areas have been described in a separate report (AWC 1987).

Please provide
this report

EXHIBIT K - CLIMATE

1.0 INTRODUCTION

The area known as Dowe Flats is located in the eastern foothills of the Rocky Mountains (elevation 5340') about three km east of Lyons, Colorado. In general, the area can be classified as having a continental climate. Being far removed from any moisture sources, the area experiences an abundance of sunny weather. The relative lack of atmospheric moisture often results in large diurnal temperature cycles. Large annual temperature cycles result from seasonal variations in solar elevation angle as well as a lack of atmospheric moisture. These characteristics are somewhat modified by the proximity of the Rocky Mountains. The mountains act as a thunderstorm growth region during the spring and summer so that Dowe Flats may receive more precipitation than a typical continental location. Mountain Chinook wind events often influence diurnal temperature differences during the winter and many winter-time daily maximum temperatures above 60°F have been recorded. Also, the gradual increase in terrain elevations near the mountains tend to block the area from shallow arctic air masses which often penetrate the plains region to the east during the winter.

Specific weather data for a climatologically significant period of record are not available for Dowe Flats. However, a climatological record is available for the Waterdale substation located approximately 17 km to the north of Dowe Flats. Waterdale occupies a similar foothills location (elevation 5230') and is near enough that the data should be applicable for a climatological evaluation of Dowe Flats. The period of record at Waterdale is 1931 - 1989 and includes temperature, precipitation and freeze data. These parameters, as well as wind speed and direction, will be discussed further in the following sections.

2.0 TEMPERATURE DATA

Table 1 lists several temperature statistics for the Waterdale station including monthly mean maximum, monthly mean minimum, monthly mean average, monthly highest and monthly lowest temperatures. The extreme high and low temperatures on record are 104°F and -34°F respectively. The annual mean maximum temperature is 63.6°F, while the annual mean minimum is 33.0°F. The annual average temperature is 48.3°F. On average, the warmest month is July, while the coldest is January.

3.0 FREEZE DATA

Freezing temperatures may occur ten months out of the year. *July and August are the only months never to record a temperature below 32°F.* Annually, the area experiences about 22 days on which the maximum temperature does not exceed freezing. The minimum temperature reaches 32°F or less about 172 days per year, on average. The length of the growing season is moderate. Table 2 lists the probable dates of the last spring freeze and the first fall freeze. There is a 50% probability that the last spring freeze will occur on or before May 14 and that the first fall freeze will occur on or after September 27.

This translates into a 50% probability that the growing season will be 136 days long.

4.0 PRECIPITATION DATA

Monthly average total precipitation and monthly average snowfall are presented in Table 3. It can be seen that the precipitation cycle is unimodal and that the maximum occurs in May and the minimum occurs in January. However, relatively high precipitation values are present through the summer and into September. This is mainly a result of thunderstorm rain showers which develop over the mountains and travel eastward during the late afternoons. Maximum snowfall occurs in March. This is the time of the year that synoptic scale storms most often follow a storm track across southern Colorado. The combination of the storm track, time of year and location of the Rocky Mountains often produce heavy wet snowfalls. Annual average total precipitation is 15.48" while annual average snowfall is 49.3".

5.0 WIND DATA

A climatological record of wind speed and direction is not available for either Dowe Flats or Waterdale. The only data available for the region has been collected during the last three calendar quarters of 1990 by McVehil-Monnett Associates at the Dowe Flats aerometric monitoring station. Since wind direction is highly influenced by local topography, the use of this short-term data set may be more appropriate than using data of a longer period of record from a distant station although it is recognized that the data here are not climatologically significant.

The windrose (Figure 1) and Table 4 represent the available wind data. It can be seen that the winds blow from either the northwest or southeast the majority of the time. Winds from the west tend to be the strongest, averaging 12.5 mph. The overall hourly average wind speed is 7.2 mph. The strongest hourly average wind speed on record is 45 mph.

TABLE 1
TEMPERATURE STATISTICS
(Waterdale Substation, Colorado)

	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
Monthly Mean Maximum (°F)	42.7	45.5	51.5	61.1	69.8	80.0	87.4	85.4	76.8	65.7	52.2	45.1
Monthly Mean Minimum (°F)	13.5	16.7	22.8	31.5	40.5	48.0	53.6	52.1	43.8	33.9	22.9	16.5
Monthly Mean Average (°F)	28.1	31.1	37.2	46.3	55.1	64.1	70.5	68.8	60.4	49.8	37.5	30.8
Highest (°F)	72	78	81	86	93	100	104	101	97	88	77	*
Lowest (°F)	-31	-34	-22	-9	17	28	37	33	18	7	-11	-26

*Value Unknown

Source: Colorado Climate Center

TABLE 2

**PROBABLE FIRST AND LAST FREEZE DATES
AND LENGTH OF GROWING SEASON*
(Waterdale Substation, Colorado)**

	Probability								
	10%	20%	30%	40%	50%	60%	70%	80%	90%
Date of Last Spring Frost (mo/day)	4/29	5/04	5/08	5/11	5/14	5/17	5/20	5/23	5/28
Date of First Fall Frost (mo/day)	9/13	9/18	9/21	9/24	9/27	9/30	10/20	10/06	10/11
Length of Growing Season (days)	115	122	127	132	136	140	145	150	157

* Based on 34 years of data ending 1987.

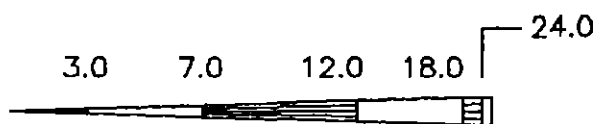
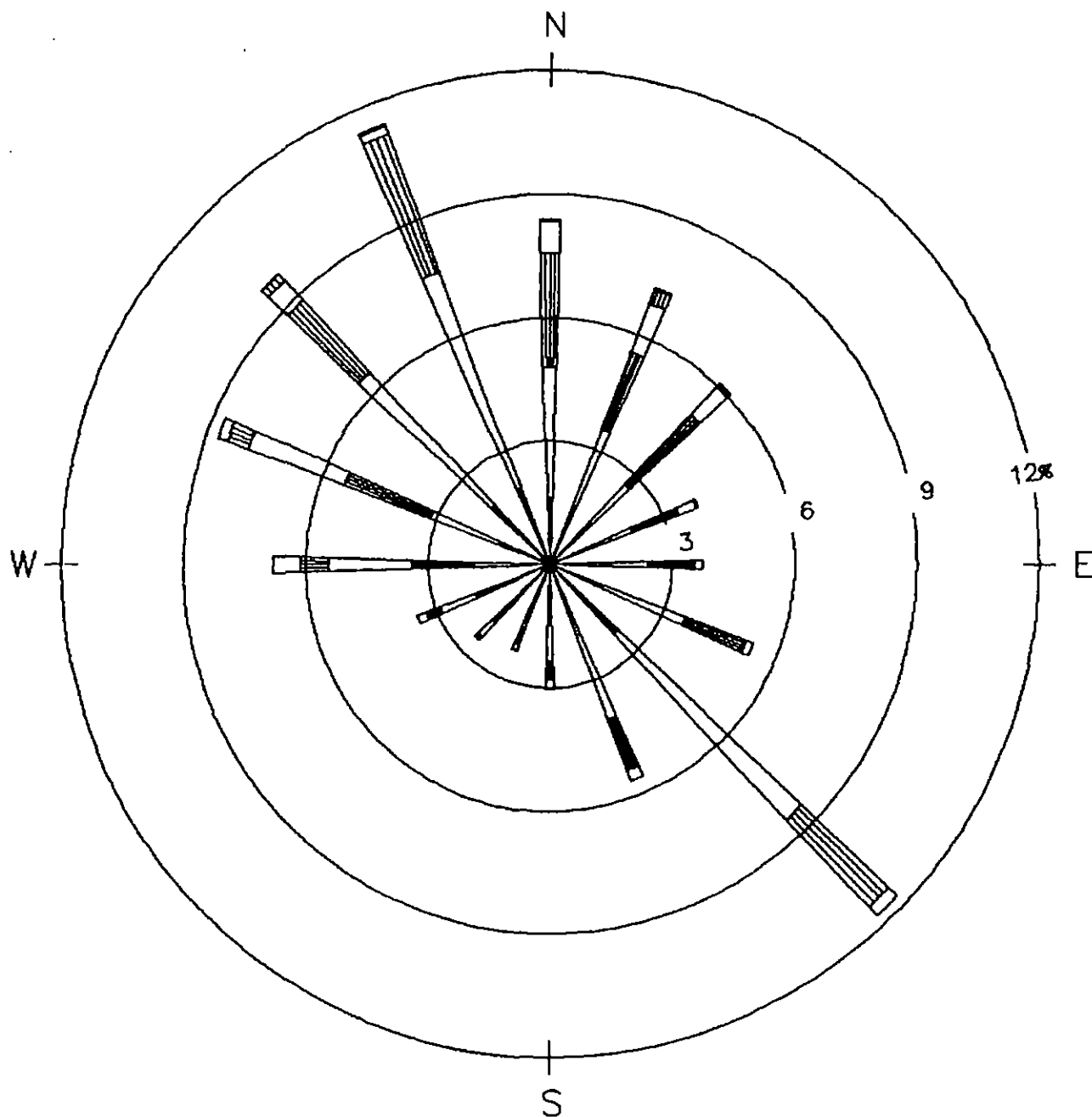
Source: Colorado Climate Center

TABLE 3

**MONTHLY TOTAL PRECIPITATION AND SNOWFALL
(Waterdale Substation, Colorado)**

	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
Monthly Total Precipitation (inches)	0.44	0.52	1.16	1.95	2.85	1.93	1.62	1.38	1.52	1.08	0.61	0.49
Monthly Snowfall (inches)	6.9	6.8	10.5	6.5	0.5	0.0	0.0	0.0	0.5	2.4	5.8	8.2

Source: Colorado Climate Center



WIND SPEED CLASS BOUNDARIES
(MILES/HOUR)

NOTES:
 DIAGRAM OF THE FREQUENCY OF
 OCCURRENCE FOR EACH WIND DIRECTION.
 WIND DIRECTION IS THE DIRECTION
 FROM WHICH THE WIND IS BLOWING.
 EXAMPLE - WIND IS BLOWING FROM THE
 NORTH 8.4 PERCENT OF THE TIME.

Source: McVehil-Monnett

WINDROSE

Dowe Flats, CO
 PERIOD: 1990

Figure 1

FROM 91/90 THROUGH 365/90

Dowe Flats

FREQUENCY OF OCCURENCE OF WIND SPEED BY DIRECTION

DIR	+-----WIND SPEED CLASSES (MPH)-----+							AVERAGE WIND SPEED
	0.- 3.	3.- 7.	7.-12.	12.-18.	18.-24.	>24.	TOTAL	
N	.0162	.0318	.0278	.0076	.0005	.0000	.0849	6.6
NNE	.0154	.0200	.0202	.0125	.0037	.0003	.0722	8.0
NE	.0104	.0162	.0240	.0092	.0011	.0000	.0609	7.7
ENE	.0099	.0119	.0119	.0047	.0003	.0000	.0388	6.6
E	.0109	.0133	.0113	.0018	.0000	.0000	.0373	5.5
ESE	.0136	.0220	.0162	.0017	.0000	.0000	.0535	5.6
SE	.0231	.0619	.0295	.0029	.0000	.0000	.1175	5.6
SSE	.0127	.0275	.0133	.0028	.0000	.0000	.0563	5.6
S	.0112	.0122	.0046	.0020	.0002	.0000	.0301	5.1
SSW	.0076	.0090	.0041	.0015	.0002	.0000	.0225	5.4
SW	.0070	.0063	.0057	.0050	.0011	.0000	.0251	7.4
WSW	.0061	.0050	.0076	.0092	.0037	.0024	.0349	11.3
W	.0075	.0110	.0154	.0206	.0070	.0066	.0682	12.5
WNW	.0122	.0194	.0226	.0251	.0054	.0024	.0872	10.1
NW	.0199	.0445	.0239	.0067	.0021	.0002	.0973	6.4
NNW	.0199	.0564	.0353	.0021	.0002	.0003	.1143	5.9
TOTAL	.2037	.3695	.2736	.1156	.0252	.0122	1.000	
AVR	2.2	5.1	8.7	14.0	20.5	28.8		7.2

CALMS AND VARIABLES FOR THIS TABLE => 43
 (CALMS AND VARIABLE DIRECTIONS ARE COUNTED IN AVERAGE
 WIND SPEED BUT NOT IN FREQUENCY COUNTS)

TOTAL NUMBER OF VALID READINGS FOR THIS TABLE => 6581
 OUT OF NUMBER READ => 6600
 ("VALID" INCLUDES CALMS)

Source: McVehil-Monnett

EXHIBIT L - RECLAMATION COSTS

Approximately 695 acres of land will be disturbed by mining and reclamation over the 25 year life of the mine. Approximately 14,300,000 cubic yards of overburden/mine waste will be placed on the reclaimed surface. This results in an average of approximately 20,575 cubic yards of overburden/mine waste per acre. This average will not apply to all portions of the reclaimed land because the actual thickness of mine waste placement varies from 0 feet up to approximately 50 feet, and the mine waste is placed in an irregular fashion.

Need
specific
phases
w/
volumes

Following the transport, placement and grading of mine waste, topsoil is replaced and the land is revegetated. Different vegetative assemblages are planned for different portions of the reclaimed area. Thus reclamation costs vary. The following table estimates reclamation costs on a per acre basis according to the planned vegetative assemblage.

of
overburden
- mine
waste
to be
replaced,
topsoil
to be
replaced

Shallow Soil (Dry) Grassland

Topsoiling and Revegetation*

Drill seeding	\$2,268/acre
Broadcast seeding	\$3,026/acre

* Please
calculate
reclamation
costs per phase

Mine Waste Placement and Grading

$20,575 \text{ yds}^3/\text{acre} \times \$0.40/\text{yd}^3 = \$8,230/\text{acre}$

} Higher for DMG to contract
The phase w/ largest
cost will be amount
of bond held for a
phased 112 operation

TOTAL PER ACRE

DRILL SEEDED	\$10,498/acre
BROADCAST SEEDED	\$11,236/acre

* This category includes:

- transporting and placing topsoil
- seedbed preparation and planting
- seed and plant costs
- organic amendments and fertilizer
- mulching

** This category includes:

- transporting mine waste

- placement
- grading and shaping
- drainage control

Shallow Soil Grassland with Pines

Topsoiling and Revegetation

Drillseeding \$2,768
Broadcast seeding \$3,526

Mine Waste and Grading

\$8230/acre

TOTAL PER ACRE

DRILL SEEDING \$10,998/acre
BROADCAST SEEDING \$11,756/acre

Deep Soil (Moist) Grassland

Topsoiling and Revegetation

\$2,962/acre

Mine Waste Placement and Grading

\$8,230/acre

TOTAL PER ACRE \$11,192

Toe Slope Grassland

Topsoiling and Revegetation

\$3,147/acre

Mine Waste Placement and Grading

\$8,230/acre

TOTAL PER ACRE \$11,377/acre

Dry Shrubland

Topsoiling and Revegetation

Drill seeding \$2,357/acre

Broadcast seeding \$2,921/acre

Irrigation (first 3 years) \$800/acre

Mine Waste Placement and Grading

\$8,230

TOTAL PER ACRE

DRILL SEEDED \$11,387/acre

BROADCAST SEEDED \$11,951/acre

Moist Shrubland

Topsoiling and Revegetation

Drill seeding \$1,710/acre

Broadcast seeding \$2,161/acre

Mine Waste Placement and Grading

\$8,230/acre

TOTAL PER ACRE

DRILL SEEDED \$9,940/acre

BROADCAST SEEDED \$10,391/acre

Herbaceous Wetland

Topsoiling and Revegetation

\$2,163/acre

Mine Waste Placement and Grading

\$8,230/acre

TOTAL PER ACRE = \$10,393

Shrub Wetland

Topsoiling and Revegetation

Drill seeding \$2,313/acre

Broadcast seeding \$2,863/acre

Mine Waste Placement and Grading

\$8,230/acre

TOTAL PER ACRE

DRILL SEEDED \$10,543/acre

BROADCAST SEEDED \$11,093/acre

Cottonwood Groves

Topsoiling and Revegetation

Drill seeding \$11,322/acre

Mine Waste Placement and Grading

\$8,230/acre

TOTAL PER ACRE = \$19,552

	Acreage	Average cost/acre	Total Cost
Shallow Soil (Dry) Grassland	160	\$10,877	\$1,740,320
Shallow Soil Grassland with Pines	49	\$11,377	\$ 557,473
Deep Soil (Moist) Grassland	10	\$11,192	\$ 111,920
Toe Slope Grassland	72	\$11,377	\$ 819,144
Dry Shrubland	227	\$10,165	\$2,307,455
Moist Shrubland	95	\$10,165	\$ 965,675
Herbaceous Wetland	33	\$10,393	\$ 342,969
Shrub Wetland	3	\$10,818	\$ 32,454
Cottonwood Groves	12	\$19,552	\$ 234,624
Open Water	44	\$ 0	\$ 0
Total	705		\$7,112,034

Average cost per acre \$10,087

It is recommended that the reclamation bond be set at \$10,000/acre.

Southdown wishes to post the bond in phases of 100 acres.

Specific phases - maps
mining plans
reclamation plans

EXHIBIT N - OTHER PERMITS AND LICENSES

The applicant is seeking the following permits:

- Special Use approval for mining and reclamation in Boulder County
- Colorado Department of Transportation access permit for access onto State Highway 66
- Air emissions permit from the Colorado Air Pollution Control Commission
- Nationwide 404 permits from the U.S Army Corps of Engineers.

EXHIBIT O - SOURCE OF LEGAL RIGHT TO ENTER

Southdown's application includes property which is divided into four parcels for purposes of this Exhibit. The parcels are called:

1. The Cotenancy Property
2. The Harroun Property
3. The Peiker-Dickens Property
4. The Frontier Materials Property

1. The Cotenancy Property

The Cotenancy Property occupies most of Dowe Flats in Sections 9, 10, 15, 16 and 21. It is so named because it was formerly owned by Southdown, Inc. and a Colorado partnership called Marigold 41. As a result of recent litigation, and subject to a Term Deed dated March 16, 1993 as recorded March 25, 1993 on film 1808, Reception No. 01276718, Boulder County Clerk and Recorder, this property is owned in fee simple by Southdown, Inc. (see Title Report).

2. The Harroun Property

The Harroun Property is in the south-central portion of the permit area in Sections 15, 16, 21 and 22. It is so named because it is owned by Michael R., Steven D., and Linda B. Harroun. Southdown, Inc. has the legal right to enter and conduct mining operations on this property subject to its right per a Memorandum of Option dated April 11, 1989, Easement Agreement, and Covenant as recorded on April 13, 1989, Boulder County Clerk and Recorder, film no. 1574, Reception numbers: 00977245, 00977246, and 00977247 respectively (see Title Report).

3. The Peiker-Dickens Property

This property is in the southeast corner of the permit area in Section 22 and is owned in fee simple by Southdown, Inc. (see Title Report).

4. The Frontier Materials, Inc. Property

This property is south of Dowe Flats and north of the cement plant property in Section 21. The mine haul road crosses this property which is owned by Frontier Materials, Inc. subject to an interest by Western Mobile, Inc. Southdown's right to use this property for a haul road is demonstrated by the attached document.

EXHIBIT P - OWNERS OF RECORD OF SURFACE AND SUBSURFACE RIGHTS

Surface Owners Within the Permit Area

Southdown, Inc.
(dba Southwestern Portland Cement Company)
5134 Ute Highway
Lyons, CO 80540

Michael R., Steven D., and Linda B. Harroun
13212 North 53rd St.
Longmont, CO 80501

Frontier Materials, Inc.
1800 Pike Road
Longmont, CO 80501

Subsurface (Limestone and Shale) Owners of the Substance to be Mined

Southdown, Inc.
(dba Southwestern Portland Cement Company)
5134 Ute Highway
Lyons, CO 80540

Michael R., Steven D., and Linda B. Harroun
13212 North 53rd Street
Longmont, CO 80501

Subject to Southdown, Inc.'s rights per (4/11/89) Memorandum of Option, Easement Agreement, and Covenant as recorded on April 13, 1989, Boulder County Clerk and Recorder, Film No. 1574 Reception Nos. 00977245, 00977246, and 00977247 respectively.

EXHIBIT Q - MUNICIPALITIES WITHIN TWO MILES

Town of Lyons
P.O. Box 49
Lyons, CO 80540

**EXHIBIT R - PROOF OF MAILING OF NOTICES TO COUNTY COMMISSIONERS
AND SOIL CONSERVATION DISTRICT**

Is your RETURN ADDRESS completed on the reverse side?

SENDER:

- Complete items 1 and/or 2 for additional services.
- Complete items 3, and 4a & b.
- Print your name and address on the reverse of this form so that we can return this card to you.
- Attach this form to the front of the mailpiece, or on the back if space does not permit.
- Write "Return Receipt Requested" on the mailpiece below the article number.
- The Return Receipt will show to whom the article was delivered and the date delivered.

I also wish to receive the following services (for an extra fee):

1. ☐ Addressee's Address
2. ☐ Restricted Delivery

Consult postmaster for fee.

3. Article Addressed to:

Boulder County Board of County
Commissioners
P.O. Box 471
Boulder, CO 80306

4a. Article Number

P433387 568

4b. Service Type

- ☐ Registered ☐ Insured
☒ Certified ☐ COD
☐ Express Mail ☐ Return Receipt for Merchandise

7. Date of Delivery

MAY 28 1993

5. Signature (Addressee)

6. Signature (Agent)

John L. Johnson

8. Addressee's Address (Only if requested and fee is paid)

PS Form 3811, December 1991 ☆ U.S.G.P.O. : 1992-307-530

DOMESTIC RETURN RECEIPT

Thank you for using Return Receipt Service.

Is your RETURN ADDRESS completed on the reverse side?

SENDER:

- Complete items 1 and/or 2 for additional services.
- Complete items 3, and 4a & b.
- Print your name and address on the reverse of this form so that we can return this card to you.
- Attach this form to the front of the mailpiece, or on the back if space does not permit.
- Write "Return Receipt Requested" on the mailpiece below the article number.
- The Return Receipt will show to whom the article was delivered and the date delivered.

I also wish to receive the following services (for an extra fee):

1. ☐ Addressee's Address
2. ☐ Restricted Delivery

Consult postmaster for fee.

3. Article Addressed to:

Longmont Soil Conservation Dist.
Box D. 9595 Nelson Rd.
Longmont, CO 80501

4a. Article Number

P433387 569

4b. Service Type

- ☐ Registered ☐ Insured
☒ Certified ☐ COD
☐ Express Mail ☐ Return Receipt for Merchandise

7. Date of Delivery

5-27-93

5. Signature (Addressee)

6. Signature (Agent)

James L. Johnson

8. Addressee's Address (Only if requested and fee is paid)

PS Form 3811, December 1991 ☆ U.S.G.P.O. : 1992-307-530

DOMESTIC RETURN RECEIPT

Thank you for using Return Receipt Service.

NOTICE OF FILING APPLICATION
FOR COLORADO MINED LAND RECLAMATION PERMIT
FOR REGULAR (112) OPERATION

NOTICE TO THE BOARD OF COUNTY COMMISSIONERS
Boulder COUNTY

Southdown, Inc. dba Southwestern

Portland Cement (Operator) has applied for a reclamation permit from the Colorado Mined Land Reclamation Board ("the Board") to conduct mining operations in Boulder County. The attached information is being provided to notify you of the location and nature of the proposed operation. The entire application is on file with the Colorado Mined Land Reclamation Division ("the Division") and the local county clerk and recorder.

The applicant proposes to reclaim the affected land to wildlife habitat. Pursuant to C.R.S. 34-32-116(7)(j) the Board is required to confer with the local Board of County Commissioners before approving of the post-mining land use. Accordingly, the Board would appreciate your comments on the proposed operation. Please note that, in order to preserve your right to a hearing before the Board on this application, you must submit written comments on the application within twenty (20) days of the date of last publication of notice pursuant to C.R.S. 34-32-112(10).

If you would like to discuss the proposed post-mining land use, or any other issue regarding this application, please contact the Colorado Mined Land Reclamation Division, 1313 Sherman Street, Room 215, Denver, Colorado 80203, (303) 866-3567.

NOTE TO OPERATOR: You must attach a copy of the first four pages of the application to this notice. If this is a notice of a change to a previously filed application you must either attach a copy of the changes, or attach a complete and accurate description of the change.

NOTICE OF FILING APPLICATION
FOR COLORADO MINED LAND RECLAMATION PERMIT
FOR REGULAR (112) OPERATION

NOTICE TO THE BOARD OF SUPERVISORS
OF THE LOCAL SOIL CONSERVATION DISTRICT
Longmont Soil Conservation DISTRICT

Southdown, Inc. dba Southwestern

Portland Cement (Operator) has applied for a reclamation permit from the Colorado Mined Land Reclamation Board ("the Board") to conduct mining operations in Boulder County. The attached information is being provided to notify you of the location and nature of the proposed operation. The entire application is on file with the Colorado Mined Land Reclamation Division ("the Division") and the local county clerk and recorder.

The applicant proposes to reclaim the affected land to wildlife habitat. Pursuant to C.R.S. 34-32-116(7)(j) the Board is required to confer with the local Soil Conservation Districts before approving of the post-mining land use. Accordingly, the Board would appreciate your comments on the proposed operation. Please note that, in order to preserve your right to a hearing before the Board on this application, you must submit written comments on the application within twenty (20) days of the date of last publication of notice pursuant to C.R.S. 34-32-112(10).

If you would like to discuss the proposed post-mining land use, or any other issue regarding this application, please contact the Colorado Mined Land Reclamation Division, 1313 Sherman Street, Room 215, Denver, Colorado 80203, (303) 866-3567.

NOTE TO OPERATOR: You must attach a copy of the first two pages of the application to this notice. If this is a notice of a change to a previously filed application you must either attach a copy of the changes, or attach a complete and accurate description of the change.

EXHIBIT S - PROOF OF FILING WITH COUNTY CLERK AND RECORDER